TRACE LAB
Trustworthy AI, Reasoning Algorithms, and Causal lEarning

Causality and Large Language Models: A Survey of System Design, Capabilities, Trustworthiness, and Applications

Samar Kale[1] **Vineeth Bekal**[1] **Aman Chadha**[2*] **Raha Moraffah**[1]

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Abstract

Large language models (LLMs) have advanced rapidly in recent years, demonstrating capabilities in reasoning, generation, and planning across diverse applications. Causality, a fundamental aspect of human intelligence, enables reasoning about cause-and-effect relationships and has contributed to advances in machine learning. Recent research has increasingly explored the use of causal methods to improve LLM capabilities, system design, and trustworthiness. However, existing studies span diverse methods and tasks, highlighting the need for a systematic understanding of how causality contributes to different aspects of LLMs. In this survey, we examine the role of causality in four main aspects of LLMs: system design, capabilities, trustworthiness, and applications. We introduce a taxonomy to organize existing approaches and examine how causal principles are used across these directions. We further review evaluation methods for all four aspects from a causal perspective. Finally, we identify key challenges and open research directions to guide future research on causality for LLMs.

1 Introduction

Large language models (LLMs) have advanced rapidly in recent years, demonstrating strong performance across a wide range of applications (Achiam et al., 2023; Grattafiori et al., 2024). Despite these advances, LLMs continue to face challenges in generation, reasoning, planning, and reliable deployment across real-world applications and downstream tasks due to their reliance on spurious correlations (Bynum and Cho, 2025; Han et al., 2025; Zan et al., 2026; Chai et al., 2025; Lyu et al., 2024).

Causality is a fundamental aspect of human intelligence that enables us to understand cause-and-effect relationships and reason about hypothetical interventions (Lake et al., 2017). Causality has long contributed to advances in machine learning Kaddour et al. (2025). Motivated by these advances, researchers have increasingly explored the integration of causal principles into LLMs (e.g., Chernozhukov et al. (2018); Guo et al. (2023); Nguyen et al. (2025)). LLM research spans four main aspects: system design, capabilities, trustworthiness, and applications. System design focuses on model development and LLM-based architectures; capabilities include generation, reasoning, planning, and decision-making; trustworthiness concerns fairness, faithfulness, interpretability, security, and alignment to ensure reliable deployment in real-world, high-stakes applications; and applications focus on the use of LLMs in downstream tasks. These aspects motivate a systematic examination of how causality can contribute to different areas of LLM research.

In this survey, we present a systematic overview of research at the intersection of causality and LLMs. We examine the use of causality across four main aspects of LLMs: system design (Section 2), capabilities (Section 3), trustworthiness (Section 4), and applications (Section 5). We introduce a taxonomy that organizes existing approaches according to their tasks, subtasks, and methods. Figure 1 presents an overview of the proposed taxonomy, and a more detailed version is presented in Appendix B. We further review evaluation

[1]{skale, vbekal, rmoraffah}@wpi.edu
[2]aman@amanchadha.com,
*Work done outside position at DeepMind

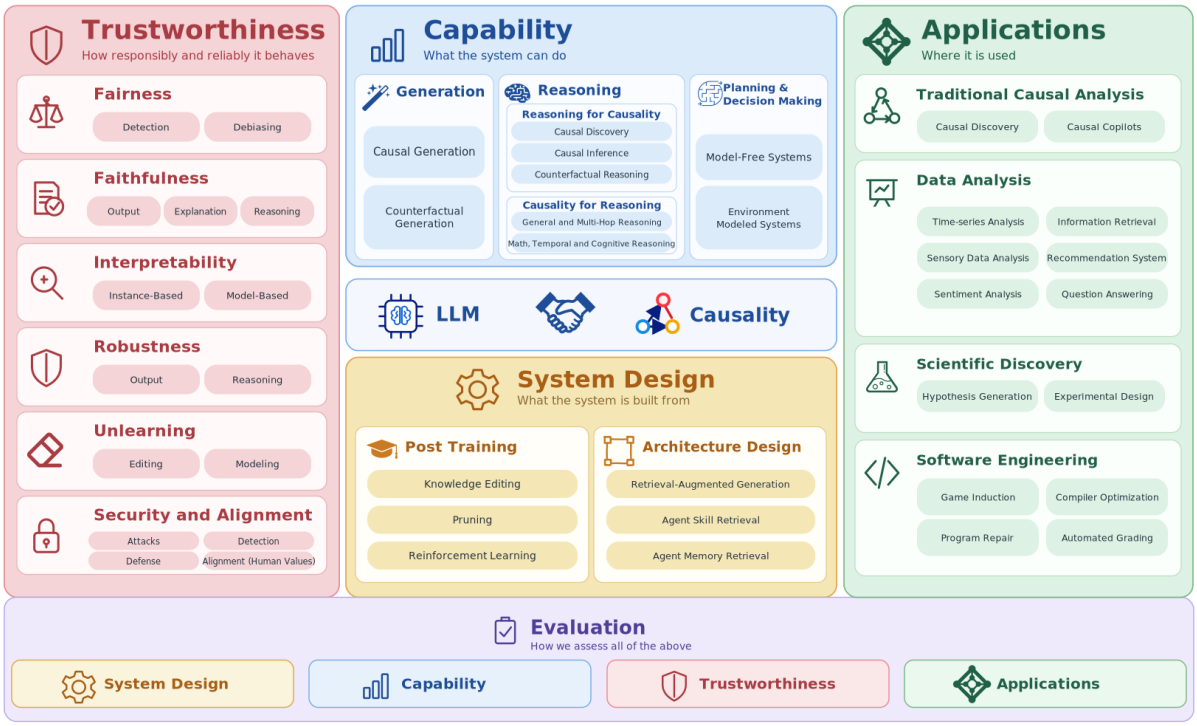


Figure 1: Overview of the proposed taxonomy, examining the role of causality across four main aspects of LLMs: system design, capabilities, trustworthiness, and applications, along with causal evaluation methods across these aspects.

methods for all four aspects from a causal perspective (Section 6). Finally, we discuss key challenges and open research directions at the intersection of causality and LLMs (Section 7).

Differences from Existing Surveys. The survey most closely related to ours is (Liu et al., 2025c), which examines both the use of LLMs for causal tasks and the use of causality to improve LLM capabilities and trustworthiness. However, its coverage of causal methods for LLM system design, downstream applications, and evaluation is limited. Other surveys focus on specific directions. (Zhang et al., 2023) discuss the feasibility and opportunities of integrating causality and LLMs, whereas (Wu et al., 2024a; Frauen et al., 2026) examine causal methods across LLM development and evaluation. Several surveys focus on specific causal capabilities, including commonsense causal reasoning (Cui et al., 2024), causal inference (Ma, 2025), causal reasoning and evaluation (Yu et al., 2025a), causal uncertainty (Cui et al., 2025), and causal reasoning enhancement (Li et al., 2025c). Other studies examine LLMs for causal discovery (Wan et al., 2025) and causal multi-agent systems (Bazgir et al., 2025). In contrast, our survey systematically examines how causality contributes to four main aspects of LLM research: system design, capabilities, trustworthiness, and applications. We introduce a taxonomy that organizes existing studies by tasks, subtasks, and methods, and examine causal evaluation strategies across these aspects. A detailed comparison with existing surveys is presented in Table 2, Appendix C.

2 Causality and System Design

In this section, we discuss causal post-training methods for improving LLM performance, as well as the use of causality in designing more complex LLM systems.

2.1 Post Training

LLM architectures can be enhanced through post-training methods to refine model's knowledge, behavior, and efficiency (Ouyang et al., 2022; Meng et al., 2022; Frantar and Alistarh, 2023; Wang et al., 2020). We

discuss how causal approaches improve such training paradigms.

Reinforcement learning (RL) traditional RL methods such as GRPO (Shao et al., 2024) treat each candidate response as conditionally independent, overlooking semantic dependencies or interactions among the candidates. Modeling the relations between the input query, candidate responses, and final outcome with a Structural Causal Model (SCM), GCPO guides the policy updates towards responses that are individually useful and structurally coherent with the group (Gu et al., 2026).

Knowledge editing selectively updates model components without full-retraining (Meng et al., 2022). To overcome the epistemic dissonance, which occurs due to isolated fact injection and recursive error accumulation, (Li et al., 2026f) propose to distill the causal reasoning trajectories with knowledge transition with asymmetric on-policy distillation. (Meng et al., 2022) identify critical layers with causal tracing and adaptively update them based on their causal efficacy, reducing the recursive error accumulation.

Pruning removes redundant or unnecessary model parameters (Frantar and Alistarh, 2023) to improve model efficiency while preserving performance. However, correlations among parameters may cause conventional pruning methods to remove causally important parameters. To address this, (Sheth et al., 2026a) use causal analysis to identify and preserve parameters that have a causal effect on model predictions, even when they are correlated with less relevant parameters. In another line of research, (Guo et al., 2025c) propose identifying tokens that confound the causal relationship between instructions and responses by comparing their gradients with those from a stronger teacher model. These tokens are then pruned to reduce spurious correlations in long-context reasoning.

2.2 Architecture Design

To extend the capabilities of LLMs and address their limitations in complex tasks, LLM-based architectures like retrieval-augmented generation and LLM agents have been developed. We discuss the role of causality in improving these architectures.

Retrieval-Augmented Generation (RAG) enhance LLMs performance by providing relevant external information for informed generation (Lewis et al., 2020), although similarity-based retrieval may retrieve causally irrelevant information. CAUSALRAG uses document-level causal graphs and discovers causal paths among the nodes to guide the retrieval process (Wang et al., 2025a), while CAUSALRAG2 (Wang et al., 2026b) extends this to hierarchical knowledge graphs with causal gates to filter for required documents causally. To reduce graph construction costs HCG-RAG (Saouda et al., 2026) constructs an outcome-oriented causal graph that decreases the required number of nodes which improves the graph construction efficiency while preserving causal relationships relevant to the given query.

LLM Agents use memory and tools for complex tasks (Park et al., 2023; Wang et al., 2024). Similarity-based memory retrieval methods can overlook contextual relevance, leading to irrelevant or misleading information. CAUSAL MEMORY INTERVENTION (Srivastava, 2026) estimates the causal effect of retrieved memories on task performance by comparing no-memory, with-memory, and perturbed-memory conditions, and filters out irrelevant or harmful memories. ACTMEM (Zhang et al., 2026b) represents conversation history as a causal graph and uses counterfactual reasoning to predict the consequences of retrieved facts, enabling the retrieval of causally relevant information. Traditional skill-retrieval methods often require long contexts, treat skills independently, or rely on accurate relevance estimates. To address these limitations, CASKG (Li et al., 2026h) constructs a directed skill graph and refines its edges through task-specific counterfactual perturbations. This enables agents to retrieve skills while preserving their procedural dependencies and execution order.

3 Causality and Capability

LLM capabilities refer to the ability of the models to perform tasks across different settings using reasoning, generation, planning, and decision-making. We discuss how causality is utilized to improve three major LLM capabilities: generation, reasoning, and planning & decision-making.

3.1 Generation

LLM generation refers to the ability to produce coherent outputs based on a given input (Brown et al., 2020). We categorize the synergy of LLM generation and causality as Causal Generation and Counterfactual Generation.

3.1.1 Counterfactual Generation

Counterfactual generation produces alternative versions of observed scenarios by intervening in specific conditions while preserving the relevant context and altering outcomes (Pearl, 2009).
Prompting: Prompt-based methods generate counterfactuals through targeted textual span extraction (Chen et al., 2023) or neuromorphic-inspired pattern separation and completion to improve commonsense consistency (Miao et al., 2024).
Training: Training-based approaches improve generation quality through counterfactual data augmentation and preference optimization (Zhang et al., 2025b) or classifier-guided selection of effective label-flipping counterfactuals (Nguyen et al., 2026a).
Architecture: At the architecture level, SCMs with Gumbel-Max distributions (Gumbel, 1954) enable token-level counterfactual generation conditioned on factual LLM outputs (Chatzi et al., 2024; Ravfogel et al., 2025), although robustness and faithfulness remain challenging.

3.1.2 Causal Generation

Causal generation refers to the ability of LLMs to generate outputs under a causal model and specific interventions that propagate the causal effects to downstream generation (Bynum and Cho, 2025). For example, SD-SCM (Bynum and Cho, 2025) combines user-provided DAG with LLM to generate data consistent with user-specified DAG.

3.2 Reasoning

LLM reasoning refers to the ability to process information and draw conclusions through logical steps. We categorize its synergy with causality into Reasoning for Causality (Section 3.2.1) and Causality for Reasoning (Section 3.2.2).

3.2.1 Reasoning for Causality

Based on the ability of LLM to perform reasoning, studies have also investigated the ability of LLM to perform causal reasoning (Kiciman et al., 2024; Chen et al., 2024a; Gao et al., 2023).
Causal Discovery identifies causal relationships among variables from observational data, to recover the underlying causal structure as a DAG or an SCM (Kiciman et al., 2024).
Prompting methods infer pairwise causal relationships (Kiciman et al., 2024), with extensions improving consistency through triplet-based causal ordering (Vashishtha et al., 2025), reducing query complexity from quadratic to linear through breadth-first search (Jiralerspong et al., 2024), and incorporating causal adjacency matrices with collider detection (Bagheri et al., 2025) or structured knowledge graphs (Sun et al., 2025a) to guide causal discovery.
Training approaches improve out-of-distribution (OOD) generalization by incorporating human-derived causal priors into attention mechanisms through Re-Attention (Han et al., 2025).

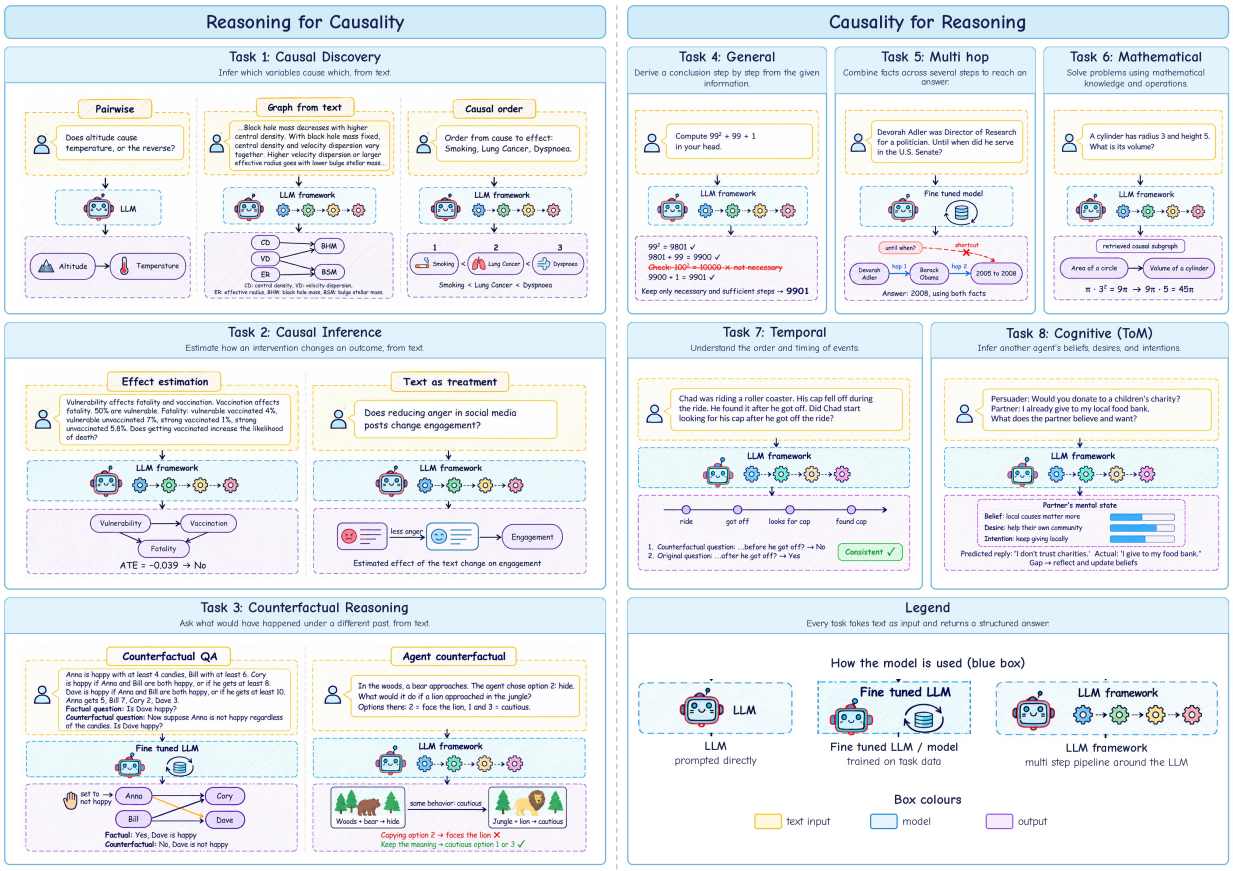


Figure 2: Illustration of tasks in reasoning for causality using LLMs.

Causal Inference estimates the effect of intervening on a variable X (cause or treatment) on an outcome Y (effect). A fundamental task in causal inference is causal effect estimation, which quantifies the effect of interventions on outcomes.

Prompting approaches guide causal effect reasoning through structured chain-of-thought (CoT) (Jin et al., 2023) or improve auditability using target-aware causal graphs, causal chain verification, and evidence aggregation (Lan et al., 2026)

Training methods address high-dimensional textual data through LLM-based nuisance estimation within the Double Machine Learning framework (Chernozhukov et al., 2018) for confounding adjustment (Veljanovski and Wood-Doughty, 2024) and LLM-generated textual interventions with domain-adversarial learning to estimate average and conditional treatment effects (Guo et al., 2025b). Alternatively, axiomatic training uses diverse synthetic causal examples to improve reasoning generalization (Vashishtha et al., 2024).
Counterfactual Reasoning reasons about the effects of hypothetical interventions against factual observations Pearl (2009).

Prompting Prompt-based approaches follow the abduction-action-prediction framework (Pearl, 2002), to infer latent factors, apply hypothetical interventions, and predict alternative outcomes (Pona et al., 2025).

Training Training-based methods improve counterfactual question answering through factual and counterfactual feedback combined with supervised fine-tuning and preference optimization to enhance generalization (Hüyük et al., 2025).

LLM Aspect	Benchmarks	Datasets
Causality and System Design	(Chen et al., 2026a)	(Luan et al., 2026; Shah, 2026)
Causality and Capability	(Jin et al., 2023; Chen et al., 2024b; Sheth et al., 2025; Yu et al., 2025a; Li et al., 2026e; Du et al., 2026; Zečević et al., 2023; Leban and Sun, 2026; Acharya et al., 2025; Chen et al., 2026a, 2024a; Xu and Fu, 2026; Chi et al., 2024; Zhang et al., 2025d; Romanou et al., 2023; Wang, 2024a; Wang et al., 2026e; Liu et al., 2024b; Sawarni et al., 2026; Sheth et al., 2024; Babakov et al., 2025; Ding et al., 2026; Chen et al., 2026c; Wu et al., 2025; Joishy et al., 2026; Sonkar et al., 2024; Nguyen et al., 2024; Ong et al., 2025; Bynum and Cho, 2025; Chen et al., 2026d)	(Miliani et al., 2025; Jin et al., 2024; Astorga et al., 2026; Liu et al., 2024b; Li et al., 2026b; Roemmele et al., 2011; Du et al., 2022; Tandon et al., 2019; Toroghi et al., 2026; González and Nori, 2024; Carro et al., 2025; Scutari, 2010a; Sachs et al., 2005; Mooij et al., 2016; Erdős and Rényi, 1959; Barabási and Albert, 1999; Frohberg and Binder, 2022; Vashishtha et al., 2026; Huang et al., 2024c)
Causality and Trustworthiness	(Xu and Fu, 2026; Liu et al., 2023c; Wang and Liu, 2026; Han et al., 2026; Adewuyi et al., 2026; Chen et al., 2026b; Mo et al., 2026; Karvonen et al., 2026; Chen et al., 2025c; Toker et al., 2026; Zaman and Srivastava, 2025; Mittal and Arike, 2026; Arora et al., 2024; Huang et al., 2024a; Mueller et al., 2025; Ye et al., 2026)	(Carro et al., 2025; Hong and Roth, 2026; Shah, 2026; Mayilvaghanan et al., 2026; Parziale et al., 2026a; Xie et al., 2026; Zhang et al., 2025e)
Causality and LLM Applications	(Leban and Sun, 2026; Acharya et al., 2025; Liu et al., 2024b; Sawarni et al., 2026; Sheth et al., 2024; Chen et al., 2026d)	(Liu et al., 2024b)

Table 1: Benchmarks and datasets for causal evaluation of large language models, grouped by the four LLM aspects in the taxonomy. Each cell cites the studies that introduced the resources.

3.2.2 Causality for Reasoning

LLMs have showcased capability of other reasoning tasks. In this section, we focus on those with the use of causality.

General and Multi-Hop Reasoning General reasoning is the ability of LLMs to analyze information and derive conclusions, while multi-hop reasoning integrates multiple pieces of evidence through intermediate reasoning steps (Jiang et al., 2022). To address irrelevant reasoning steps and shortcut dependencies in CoT reasoning (Wei et al., 2022), causal approaches estimate step importance through probabilities of necessity and sufficiency (Pearl, 2002) using counterfactual replacements, subsequently improving reasoning through fine-tuning or in-context learning (Yu et al., 2025b). Similarly, multi-hop reasoning can be modeled as a DAG, with counterfactual inference distinguishing shortcut paths from valid reasoning dependencies to guide training (Guo et al., 2023).

Mathematical, Temporal, and Cognitive Reasoning Specialized mathematical reasoning benefits from causal discovery over mathematical knowledge and question-solution pairs, with relevant causal subgraphs retrieved to guide problem-solving (Zan et al., 2026). Temporal reasoning uses counterfactual question variations and predictions aggregation to improve consistency under lexical perturbations (Kim and Hwang, 2025). For cognitive reasoning, particularly, Theory of Mind (ToM), counterfactual reflection supports reasoning about alternative mental states TOM-AGENT(Yang et al., 2025), while causal tracing and activation steering target ToM-related representations to improve social reasoning (Li et al., 2026d). Additionally, SCM-based opponent modeling captures dependencies between opponent's states and actions to improve behavioral prediction in dynamic interactions (Cao et al., 2026)

3.3 Planning & Decision Making

Planning organizes a sequences of actions to achieve a goal, while decision making selects a specific action for a goal (Yao et al., 2023b,a). Although planning & decision making are an application of reasoning capability, we discuss this as a separate section due to its importance to enable LLMs for goal-directed behavior and autonomous task execution. We divide the planning & decision-making tasks into two main

categories: Model-Free Systems and Environment Modeled Systems.

3.3.1 Model-Free Systems

Model-Free systems enable LLMs to plan and make decisions without explicitly learning or generating a predictive model of the environment dynamics, instead relying on observations and environmental feedback from the environment (Yao et al., 2023b; Shinn et al., 2023).
Decision Making: ARGORA (Jin et al., 2026) uses argumentation graphs and counterfactual inference to quantify influential factors in multi-agent decisions.
Planning: CAUSALMACE (Chai et al., 2025) uses causal inference to refine task dependencies under game and task constraints, guiding coordinated multi-agent planning.

3.3.2 Environment Modeled Systems

Environment modeled systems learn a predictive model of the environment capable of helping agents simulate the effects of actions on future states and support planning & decision making (Hao et al., 2023; Guan et al., 2023; Tang et al., 2024).
Decision Making: SCM is used to represent state-action dependencies, enabling adaptive decision making through environmental interventions and model updates (Chen et al., 2025b).
Planning: SCM-based methods support causal action dependency discovery (Nguyen et al., 2025), counterfactual refinement of actions in stochastic environments (Fu et al., 2026) are two lines of research. Additionally, causal representation learning (Schölkopf et al., 2021) enables the construction of causal world models for causally informed planning (Gkountouras et al., 2025).

4 Causality and Trustworthiness

LLM trustworthiness refers to the extent to which LLM behavior is fair, faithful, interpretable, robust and safe while aligning with human preferences and ethics (Liu et al., 2023b; Huang et al., 2024b). With the synergy of causality, the trustworthiness of LLMs can be enhanced by learning the underlying behavior of the model, mitigating spurious correlations, and enabling a causal analysis of the LLM's predictions and decisions. In this section, we discuss the role of causality for different aspect of trustworthiness.

4.1 Fairness

LLMs have demonstrated bias and reproduce stereotypes in their generated outputs (Parrish et al., 2022; Dhamala et al., 2021). For detection, counterfactual fairness testing evaluates sensitivity of the model to sensitive attributes through stereotype-aware prompt generation (Parziale et al., 2026b), while causal interventions on inputs of agentic evaluator identify sources of bias (Chen et al., 2025a). For debiasing a black-box model, a prompting framework uses *dual direction priors* along with *selective mechanisms* to mitigate biased generation (Li et al., 2025a). Training-based methods address spurious correlations and confounding through counterfactual data augmentation (Madhavan et al., 2023), interventional reinforcement learning (Xia et al., 2024), causal-contrastive learning with fairness-sensitive objectives (Feng et al., 2026), and adversarial learning of de-biased steering representations (Chu et al., 2024).

4.2 Faithfulness

Faithfulness assess whether LLM generated outputs are grounded to the source information and whether their reasoning and explanations accurately reflect model behavior (Turpin et al., 2023; Maynez et al., 2020). Recent studies focus on the faithfulness of LLMs explanations from 3 major perspectives; output, explanation, and reasoning faithfulness.
Output faithfulness For output faithfulness, counterfactual probing and iterative refinement assess sensitivity to factual perturbations (Feng, 2025; Narendra and Chatterjee, 2026), while causal graph-guided fine-tuning and causal-effect based abstention aim to mitigate hallucinations (Li et al., 2026g; Nguyen et al., 2026b).

Explanation faithfulness For explanation faithfulness, causal concept interventions quantify the influence of explanatory factors (Matton et al., 2025), while studies reveal the unreliability of self-generated counterfactual explanations (Madsen et al., 2024; Mayne et al., 2025). Alternatively, counterfactual simulatability evaluates the precision of explanations (Chen et al., 2024c; Limpijankit et al., 2025; Karvonen et al., 2026).
Reasoning faithfulness For the faithfulness of reasoning, methods improve the causal consistency of reasoning traces through interventions (Uikey, 2025), preference optimization over intervened reasoning chains Swaroop et al. (2025), multi-agent causal verification (Tang et al., 2025), and SCM guided reinforcement learning for faithful multi-hop reasoning (Bu et al., 2026).

4.3 Interpretability

Interpretability aims to explain the mechanisms behind LLMs behavior (Feldhus and Kopf, 2025; Zhang and Nanda, 2024). Causal approaches contribute to the field by identifying causally influential components in the model, distinguishing correlational associations from causal mechanisms (Meng et al., 2022).
Instance-Based Instance-based methods use counterfactual generation and augmentation to assess the prediction sensitivity and explain individual output (Han et al., 2020; Gat et al., 2024; Bhattacharjee et al., 2024; Nussbaum-Hoffer et al., 2026).
Model-Based Model-based methods use causal abstraction (Geiger et al., 2025; Sutter et al., 2025; Geiger et al., 2024; Wu et al., 2023), causal attribution (Syed et al., 2024). Alternatively, these methods focus on different levels such as feature-level (Ji et al., 2025; Bonagiri et al., 2026; Shah, 2026; Goyal et al., 2025; Wang et al., 2025c), component-level (Meng et al., 2022; Yan et al., 2026; Ortu et al., 2024), and system-level (Gupta et al., 2025a; Lyu et al., 2026; Triantafyllou et al., 2025; Hooda et al., 2024; Gupta et al., 2025b).

4.4 Robustness

Robustness is a crucial aspect for LLMs to maintain reliable performance despite external influences or distribution shifts (Wang et al., 2023; Zhu et al., 2024). In this section we discuss the robustness of responses and reasoning of LLMs.
Output robustness To enforce output robustness, training-based methods address distribution shifts and spurious correlations through SCM guided representation decomposition (Yu et al., 2026a) and causal identification of model components for targeted policy optimization (Zhou et al., 2025).
Reasoning robustness For robustness of reasoning, prompt-based methods leverage front-door-adjustment criteria (Zhang et al., 2025a; Wu et al., 2024b; Zhao et al., 2025; Li et al., 2026a), while training-based methods use counterfactual rewards (Wang et al., 2026a), counterfactual distillation of multi-view reasoning traces (Feng et al., 2024) and independent causal mechanisms that separate invariant and domain-specific representations of OOD generalization Gendron et al. (2024a).

4.5 Unlearning

Unlearning is the task of removing specific knowledge from an LLM, this is done by selectively removing the influence of data from the model while preserving its capabilities (Yao et al., 2024; Liu et al., 2024e). Causal approaches support unlearning through process modeling and model editing.
Modeling Modeling based methods rely on SCMs to mitigate confounding effects through backdoor adjustment (Liu et al., 2024d) or counterfactual reasoning traces with preference optimization to suppress targeted knowledge in large reasoning models (Li et al., 2026c).
Editing Model editing based approaches localize knowledge-bearing components through causal tracing and intervene on their activations for targeted removal (Fang et al., 2026). Another approach uses probabilities of sufficiency and necessity to guide attention-head interventions and fine-tuning to reduce LLM-toxicity (Wang et al., 2026d).

4.6 Security and Alignment

Security helps protect LLM or LLM-based systems from attacks, misuse, or other threats (Das et al., 2025), while, alignment ensures that the behavior presented by LLM is consistent with human intentions, preferences, and values (Shen et al., 2023). Causal approaches contribute to security by adversarial attacks, detection, and defense.

Attacks For attacks, front-door adjustment has been used to bypass LLM safety mechanisms through mediator-based jailbreak (Zhou et al., 2026).

Detection Detection methods use counterfactual probing to identify prompt injection vulnerabilities (Liu et al., 2025a), counterfactual information-flow analysis to trace unsafe agent propagations (Cai et al., 2026), interventional behavior mapping to detect LLM misbehavior (Zhang et al., 2025c), and causal influence diagrams to support proactive identification of harmful actions (Hahm et al., 2025).

Defense Defensive method use front-door adjustment with guard mediators to mitigate backdoor attack (Liu et al., 2024c), alongside counterfactual execution testing (He et al., 2026b), and temporal causal analysis (Zhang et al., 2026a) to prevent indirect prompt injection attacks.

Alignment For alignment, SCM based interventions enable counterfactual generation for pluralistic value alignment (Guo et al., 2025a), while causal reward modeling supports preference learning (Kobalczyk and Van Der Schaar, 2025).

5 Causality and LLM Applications

We discuss how the integration of causality and LLMs is applied to tasks in causal and data analysis, scientific discovery, and software engineering.

Traditional Causal Analysis focuses on causal reasoning using tabular or numerical data. LLMs support causal discovery by extracting variable metadata and domain knowledge (Liu et al., 2024a; Abdulaal et al., 2024; Li and Russo, 2026; Kim, 2026; Li et al., 2024a), generating causal priors (Lee et al., 2024; Kampani et al., 2024; Ban et al., 2025a,b; Baldo et al., 2026; Lee et al., 2025), or selecting between numerical methods and expert recommendations (Clivio et al., 2025). They also refine causal graphs using world knowledge (Khatibi et al., 2025; TAKAYAMA et al., 2025) or fine-tuning (Dong et al., 2025). Other studies integrate LLMs with traditional methods for iterative graph refinement (Du et al., 2025; Feng et al., 2025a), causal prior generation and refinement (Jiang et al., 2024; Ashwani et al., 2024; Dhawan et al., 2024; Sun et al., 2025b; Gendron et al., 2024b), and joint graph construction (Liu et al., 2026). For fairness-aware discovery, LLM-GUIDED CD (Zanna and Sano, 2025) combines LLM-based causal discovery (Jiralerspong et al., 2024) with active learning to identify fairness-relevant causal paths. Beyond causal discovery, LLMs serve as copilots for end-to-end causal reasoning (Berrevoets et al., 2025; Wang et al., 2025b; Verma et al., 2025) and construct executable SCMs for causal simulation (Astorga et al., 2026).

We cover the applications of data analysis, scientific discovery and software engineering in Appendix E

6 Evaluation and Benchmarking

This section presents an overview of the evaluation strategies for LLM-architecture, LLM-capability, LLM-trustworthiness, and applications. A comprehensive list of benchmarks can found in Table 1. Detailed discussion can be found in Appendix F.

The evaluation strategies mainly revolve around LLMs, in which they evaluate the causal effectiveness of post-training strategies (eg: CAUSALGYM (Arora et al., 2024)), evaluating reasoning (eg: RECITE (Saklad et al., 2026)) and generation (eg: CLOMO (Huang et al., 2024d)) aspects, trustworthiness of LLMs (eg: MIB (Mueller et al., 2025)). Whereas, the application benchmarks (eg: CAUSCIBENCH (Acharya et al., 2025)) evaluate the downstream capabilities of LLM.

7 Challenges & Future Works

Research at the intersection of causality and LLMs has advanced system design,model capabilities, and trustworthiness. Despite this progress, several challenges and open research directions persist.

Gap Between Causal Capabilities and Applications Although many LLM applications rely on causal inference and counterfactual reasoning, limited research directly addresses improving these capabilities. This gap calls for methods that strengthen causal reasoning and generalize across tasks and domains.

Causal Reliability of LLM Frameworks Many LLM-based frameworks, particularly in high-stakes applications, rely on generated explanations and reasoning whose faithfulness and causal validity are uncertain. Future research should develop causal evaluation methods to assess how these frameworks make decisions and whether their outputs are consistent with the underlying causal relationships.

Causal Validity and Generalization Existing evaluations often emphasize task accuracy on narrow or synthetic benchmarks, which may not adequately assess causal validity or generalization. Developing scalable evaluation methods across diverse tasks and real-world scenarios is an important research direction.

8 Conclusion

In this survey, we comprehensively explore the synergy of causality and LLMs. We present a detailed overview of the developing field and propose a taxonomy to highlight the various contributions through different perspectives. Although there are many developments in this field, we propose current challenges and future work to encourage the development of advanced LLMs and causality.

Limitations

This work aims to provide a comprehensive overview on the synergy of LLMs and causality. Our analysis is focused on the comprehensive breadth of LLM and causality encompassing both LLM of causality and causality for LLM perspectives hence, does not delve into the depths of internal mechanisms or implementation of causality or role of LLM in every work. Moreover, the field of causality and LLMs is a rapidly evolving research frontier where significant developments appear across NLP/ML venues and pre-prints. Given the pace and nature of this field our selection criteria prioritizes comprehensive coverage of key developments over exhaustive enumeration. For pre-prints, the authors of this paper manually reviewed them to assess their quality and relevance to the topic.

References

Ahmed Abdulaal, adamos hadjivasiliou, Nina Montana-Brown, Tiantian He, Ayodeji Ijishakin, Ivana Drobnjak, Daniel C. Castro, and Daniel C. Alexander. 2024. Causal modelling agents: Causal graph discovery through synergising metadata- and data-driven reasoning. In *The Twelfth International Conference on Learning Representations*.

Sawal Acharya, Terry Jingchen Zhang, Andrew Kim, Anahita Haghighat, Sun Xianlin, Rahul Babu Shrestha, Maximilian Mordig, Furkan Danisman, Clijo Jose, Yahang Qi, Pepijn Cobben, Bernhard Schölkopf, Mrinmaya Sachan, and Zhijing Jin. 2025. Causcibench: Assessing LLM causal reasoning for scientific research. In *NeurIPS 2025 Workshop on CauScien: Uncovering Causality in Science*.

Josh Achiam, Steven Adler, Sandhini Agarwal, Lama Ahmad, Ilge Akkaya, Florencia Leoni Aleman, Diogo Almeida, Janko Altenschmidt, Sam Altman, Shyamal Anadkat, and 1 others. 2023. Gpt-4 technical report. *arXiv preprint arXiv:2303.08774*.

Thanni Adewuyi, Anuoluwa Sotome, Samuel Okoko, Angel Ezendu, Oluwafunke Akinbuwa, Oluwaseun Odunsi, Oluwasegun Oguntuase, Ifeoma Nwabueze, and Abiodun Adereni. 2026. Mamabench: Benchmarking llm robustness in maternal and child health diagnosis through counterfactual clinical perturbation. *Preprint*, arXiv:2607.14385.

Aryaman Arora, Dan Jurafsky, and Christopher Potts. 2024. CausalGym: Benchmarking causal interpretability methods on linguistic tasks. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 14638–14663, Bangkok, Thailand. Association for Computational Linguistics.

Swagata Ashwani, Kshiteesh Hegde, Nishith Reddy Mannuru, Dushyant Singh Sengar, Mayank Jindal, Krishna Chaitanya Rao Kathala, Dishant Banga, Vinija Jain, and Aman Chadha. 2024. Cause and effect: Can large language models truly understand causality? *Proceedings of the AAAI Symposium Series*, 4(1):2–9.

Nicolás Astorga, Anita Kriz, and Mihaela van der Schaar. 2026. Causim: Scaling causal reasoning with increasingly complex causal simulators. *Preprint*, arXiv:2605.09079.

Nikolay Babakov, Ehud Reiter, and Alberto Bugarín. 2025. CausalGraphBench: a benchmark for evaluating language models capabilities of causal graph discovery. In *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 4: Student Research Workshop)*, pages 240–258.

Rahul Babu Shrestha, Simon Malberg, and Georg Groh. 2025. From causal parrots to causal prophets? towards sound causal reasoning with large language models. In *Proceedings of the 5th International Conference on Natural Language Processing for Digital Humanities*, pages 319–333, Albuquerque, USA. Association for Computational Linguistics.

Abdolmahdi Bagheri, Matin Alinejad, Mahdi Dehshiri, Kevin Bello, and Alireza Akhondi-Asl. 2025. C^2P: Featuring large language models with causal reasoning. *Preprint*, arXiv:2407.18069.

Federico Baldo, Simon Ferreira, and Charles K. Assaad. 2026. Retrieving classes of causal orders with inconsistent knowledge bases. In *Proceedings of the Fifth Conference on Causal Learning and Reasoning*, volume 323 of *Proceedings of Machine Learning Research*, pages 393–422. PMLR.

Taiyu Ban, Lyuzhou Chen, Derui Lyu, Xiangyu Wang, Qinrui Zhu, and Huanhuan Chen. 2025a. Llm-driven causal discovery via harmonized prior. *IEEE Transactions on Knowledge and Data Engineering*, 37(4):1943–1960.

Taiyu Ban, Lyuzhou Chen, Derui Lyu, Xiangyu Wang, Qinrui Zhu, and Huanhuan Chen. 2025b. Llm-driven causal discovery via harmonized prior. *IEEE Transactions on Knowledge and Data Engineering*, 37(4):1943–1960.

Guangsheng Bao, Hongbo Zhang, Cunxiang Wang, Linyi Yang, and Yue Zhang. 2025. How likely do LLMs with CoT mimic human reasoning? In *Proceedings of the 31st International Conference on Computational Linguistics*, pages 7831–7850, Abu Dhabi, UAE. Association for Computational Linguistics.

Albert-László Barabási and Réka Albert. 1999. Emergence of scaling in random networks. *Science*, 286(5439):509–512.

Adib Bazgir, Amir Habibdoust, Yuwen Zhang, and Xing Song. 2025. Causal mas: A survey of large language model architectures for discovery and effect estimation. *Preprint*, arXiv:2509.00987.

Jeroen Berrevoets, Julianna Piskorz, Robert Davis, Harry Amad, Jim Weatherall, and Mihaela van der Schaar. 2025. Technical report: Facilitating the adoption of causal inference methods through llm-empowered co-pilot. *Preprint*, arXiv:2508.10581.

Amrita Bhattacharjee, Raha Moraffah, Joshua Garland, and Huan Liu. 2024. Zero-shot llm-guided counterfactual generation: A case study on nlp model evaluation. In *2024 IEEE International Conference on Big Data (BigData)*, pages 1243–1248.

Preetom Biswas, Giulia Pedrielli, and K. Selçuk Candan. 2026. Causality by abstraction: Symbolic rule learning in multivariate timeseries with large language models. *Preprint*, arXiv:2602.17829.

Akash Bonagiri, Devang Borkar, Gerard Janno Anderias, Setareh Rafatirad, and Houman Homayoun. 2026. Causalflow: Causal attribution and counterfactual repair for llm agent failures. *Preprint*, arXiv:2605.25338.

Tom Brown, Benjamin Mann, Nick Ryder, Melanie Subbiah, Jared D Kaplan, Prafulla Dhariwal, Arvind Neelakantan, Pranav Shyam, Girish Sastry, Amanda Askell, Sandhini Agarwal, Ariel Herbert-Voss, Gretchen Krueger, Tom Henighan, Rewon Child, Aditya Ramesh, Daniel Ziegler, Jeffrey Wu, Clemens Winter, and 12 others. 2020. Language models are few-shot learners. In *Advances in Neural Information Processing Systems*, volume 33, pages 1877–1901. Curran Associates, Inc.

Yunhan Bu, quan zhang, Huaping Zhang, Guotong Geng, Chunxiao Gao, Askar Hamdulla, Juan Wang, Qiuchi Li, Baohua Zhang, Yunbo Cao, Zhunchen Luo, and Shuai Lei. 2026. Grounding multi-hop reasoning in structural causal models via group relative policy optimization. In *Forty-third International Conference on Machine Learning*.

Lucius E.J. Bynum and Kyunghyun Cho. 2025. Language models as causal effect generators. In *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing*, pages 2096–2115, Suzhou, China. Association for Computational Linguistics.

Yuandao Cai, Wensheng Tang, Cheng Wen, and Shengchao Qin. 2026. Ghost in the agent: Redefining information flow tracking for llm agents. *Preprint*, arXiv:2604.23374.

Shiyue Cao, Pei Xu, Likun Yang, Lei Cui, Xiaotang Chen, and Kaiqi Huang. 2026. Som: Structured opponent modeling for llm-based agents via structural causal model. In *Proceedings of the 25th International Conference on Autonomous Agents and Multiagent Systems*, AAMAS. International Foundation for Autonomous Agents and Multiagent Systems.

María Victoria Carro, Denise Alejandra Mester, Francisca Gauna Selasco, Giovanni Franco Gabriel Marraffini, Mario Alejandro Leiva, Gerardo I. Simari, and María Vanina Martinez. 2025. Do large language models show biases in causal learning? insights from contingency judgment. *Preprint*, arXiv:2510.13985.

Qi Chai, Zhang Zheng, Junlong Ren, Deheng Ye, Zichuan Lin, and Hao Wang. 2025. CausalMACE: Causality empowered multi-agents in Minecraft cooperative tasks. In *Findings of the Association for Computational Linguistics: EMNLP 2025*, pages 14410–14426, Suzhou, China. Association for Computational Linguistics.

Ivi Chatzi, Nina L. Corvelo Benz, Eleni Straitouri, Stratis Tsirtsis, and Manuel Gomez Rodriguez. 2024. Counterfactual token generation in large language models. In *Causality and Large Models @NeurIPS 2024*.

Junqi Chen, Sirui Chen, and Chaochao Lu. 2026a. Can post-training transform LLMs into causal reasoners? In *Findings of the Association for Computational Linguistics: ACL 2026*, pages 17020–17038, San Diego, California, United States. Association for Computational Linguistics.

Meilin Chen, Jian Tian, Liang Ma, Di Xie, Weijie Chen, and Jiang Zhu. 2025a. Unbiased evaluation of large language models from a causal perspective. In *Proceedings of the 42nd International Conference on Machine Learning*, volume 267 of *Proceedings of Machine Learning Research*, pages 9105–9120. PMLR.

Sirui Chen, Bo Peng, Meiqi Chen, Ruiqi Wang, Mengying Xu, Xingyu Zeng, Rui Zhao, Shengjie Zhao, Yu Qiao, and Chaochao Lu. 2024a. Causal evaluation of language models. *arXiv preprint arXiv:2405.00622*.

Sirui Chen, Mengying Xu, Kun Wang, Xingyu Zeng, Rui Zhao, Shengjie Zhao, and Chaochao Lu. 2024b. CLEAR: Can language models really understand causal graphs? In *Findings of the Association for Computational Linguistics: EMNLP 2024*, pages 6247–6265, Miami, Florida, USA. Association for Computational Linguistics.

Wei Chen, Jiahao Zhang, Haipeng Zhu, Boyan Xu, Zhifeng Hao, Keli Zhang, Junjian Ye, and Ruichu Cai. 2025b. Causal-aware large language models: Enhancing decision-making through learning, adapting and acting. In *Proceedings of the Thirty-Fourth International Joint Conference on Artificial Intelligence, IJCAI-25*, pages 4292–4300. International Joint Conferences on Artificial Intelligence Organization. Main Track.

Wenting Chen, Guolin Huang, Wenxuan Wang, and Zhongrui Zhu. 2026b. MedEinst: Benchmarking the einstellung effect in medical LLMs through counterfactual differential diagnosis. In *Proceedings of the 64th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 39778–39798, San Diego, California, United States. Association for Computational Linguistics.

Yan Chen and Cheng Liu. 2025. Causal inference based transfer learning with llms: An efficient framework for industrial rul prediction. *Preprint*, arXiv:2503.17686.

Yanda Chen, Ruiqi Zhong, Narutatsu Ri, Chen Zhao, He He, Jacob Steinhardt, Zhou Yu, and Kathleen Mckeown. 2024c. Do models explain themselves? Counterfactual simulatability of natural language explanations. In *Proceedings of the 41st International Conference on Machine Learning*, volume 235 of *Proceedings of Machine Learning Research*, pages 7880–7904. PMLR.

Yuefei Chen, Vivek K. Singh, Jing Ma, and Ruixiang Tang. 2026c. Counterbench: evaluating and improving counterfactual reasoning in large language models. In *Proceedings of the Fortieth AAAI Conference on Artificial Intelligence and Thirty-Eighth Conference on Innovative Applications of Artificial Intelligence and Sixteenth Symposium on Educational Advances in Artificial Intelligence*, AAAI'26/IAAI'26/EAAI'26. AAAI Press.

Yuen Chen, Vethavikashini Chithrra Raghuram, Justus Mattern, Rada Mihalcea, and Zhijing Jin. 2025c. Causally testing gender bias in LLMs: A case study on occupational bias. In *Findings of the Association for Computational Linguistics: NAACL 2025*, pages 4999–5019, Albuquerque, New Mexico. Association for Computational Linguistics.

Zeming Chen, Qiyue Gao, Antoine Bosselut, Ashish Sabharwal, and Kyle Richardson. 2023. DISCO: Distilling counterfactuals with large language models. In *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, Toronto, Canada. Association for Computational Linguistics.

Zhenhao Chen, Yongqiang Chen, Chenxi Liu, Junchi Yu, Xiangchen Song, Zijian Li, Jialin Li, Philip Torr, Bo Han, and Kun Zhang. 2026d. Causalgame: Benchmarking causal thinking of LLM agents in games. In *ICLR 2026 Workshop on Foundation Models for Science: Real-World Impact and Science-First Design*.

Victor Chernozhukov, Denis Chetverikov, Mert Demirer, Esther Duflo, Christian Hansen, Whitney Newey, and James Robins. 2018. Double/debiased machine learning for treatment and structural parameters. *The Econometrics Journal*, 21(1):C1–C68.

Haoang Chi, He Li, Wenjing Yang, Feng Liu, Long Lan, Xiaoguang Ren, Tongliang Liu, and Bo Han. 2024. Unveiling causal reasoning in large language models: Reality or mirage? In *Advances in Neural Information Processing Systems*, volume 37, pages 96640–96670. Curran Associates, Inc.

Yu Ying Chiu, Liwei Jiang, and Yejin Choi. 2025. DailyDilemmas: Revealing Value Preferences of LLMs with Quandaries of Daily Life. In *The Thirteenth International Conference on Learning Representations*.

Zhixuan Chu, Yan Wang, Longfei Li, Zhibo Wang, Zhan Qin, and Kui Ren. 2024. A causal explainable guardrails for large language models. In *Proceedings of the 2024 on ACM SIGSAC Conference on Computer and Communications Security*, CCS '24, page 1136–1150, New York, NY, USA. Association for Computing Machinery.

Oscar Clivio, Divyat Mahajan, Perouz Taslakian, Sara Magliacane, Ioannis Mitliagkas, Valentina Zantedeschi, and Alexandre Drouin. 2025. Learning to defer for causal discovery with imperfect experts. In *ICLR 2025 Workshop on Reasoning and Planning for Large Language Models*.

Shaobo Cui, Zhijing Jin, Bernhard Schölkopf, and Boi Faltings. 2024. The odyssey of commonsense causality: From foundational benchmarks to cutting-edge reasoning. In *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing*, pages 16722–16763, Miami, Florida, USA. Association for Computational Linguistics.

Shaobo Cui, Luca Mouchel, and Boi Faltings. 2025. Uncertainty in causality: A new frontier. In *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 8022–8044, Vienna, Austria. Association for Computational Linguistics.

Zhiqing Cui, Binwu Wang, Qingxiang Liu, Yeqiang Wang, Zhengyang Zhou, Yuxuan Liang, and Yang Wang. 2026. Augur: Modeling covariate causal associations in time series via large language models. In *Proceedings of the 64th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 764–787, San Diego, California, United States. Association for Computational Linguistics.

Badhan Chandra Das, M. Hadi Amini, and Yanzhao Wu. 2025. Security and privacy challenges of large language models: A survey. *ACM Comput. Surv.*, 57(6).

Zahra Dehghanighobadi, Asja Fischer, and Muhammad Bilal Zafar. 2025. Can LLMs explain themselves counterfactually? In *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing*, pages 7787–7815, Suzhou, China. Association for Computational Linguistics.

Jwala Dhamala, Tony Sun, Varun Kumar, Satyapriya Krishna, Yada Pruksachatkun, Kai-Wei Chang, and Rahul Gupta. 2021. Bold: Dataset and metrics for measuring biases in open-ended language generation. In *Proceedings of the 2021 ACM Conference on Fairness, Accountability, and Transparency*, FAccT '21, page 862–872, New York, NY, USA. Association for Computing Machinery.

Nikita Dhawan, Leonardo Cotta, Karen Ullrich, Rahul G. Krishnan, and Chris J. Maddison. 2024. End-to-end causal effect estimation from unstructured natural language data. In *Advances in Neural Information Processing Systems*, volume 37, pages 77165–77199. Curran Associates, Inc.

Ziyi Ding, Chenfei Ye-Hao, Zheyuan Wang, and Xiao-Ping Zhang. 2026. Step-by-step causality: Transparent causal discovery with multi-agent tree-query and adversarial confidence estimation. *arXiv preprint arXiv:2601.10137*.

Juncheng Dong, Yiling Liu, Ahmed Aloui, Vahid Tarokh, and David Carlson. 2025. CARE: Turning LLMs into causal reasoning expert. In *First Workshop on Foundations of Reasoning in Language Models*.

Huaming Du, Yujia Zheng, Baoyu Jing, Yu Zhao, Gang Kou, Guisong Liu, Tao Gu, Weimin Li, and Carl Yang. 2025. Causal discovery through synergizing large language model and data-driven reasoning. In *Proceedings of the 31st ACM SIGKDD Conference on Knowledge Discovery and Data Mining V.2*, KDD '25, page 543–554, New York, NY, USA. Association for Computing Machinery.

Jin Du, Li Chen, Xun Xian, An Luo, Fangqiao Tian, Ganghua Wang, Charles Doss, Xiaotong Shen, and Jie Ding. 2026. Ice cream doesn't cause drowning: Benchmarking LLMs against statistical pitfalls in causal inference. In *International Conference on Learning Representations*.

Li Du, Xiao Ding, Kai Xiong, Ting Liu, and Bing Qin. 2022. e-CARE: a new dataset for exploring explainable causal reasoning. In *Proceedings of the 60th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 432–446.

Paul Erdős and Alfréd Rényi. 1959. On random graphs I. *Publicationes Mathematicae Debrecen*, 6:290–297.

Ce Fang, Zhikun Zhang, Min Chen, Qing Liu, Lu Zhou, Zhe Liu, and Yunjun Gao. 2026. Kuda: Knowledge unlearning by deviating representation for large language models. *Preprint*, arXiv:2602.19275.

Amirmohammad Farzaneh, Salvatore D'Oro, and Osvaldo Simeone. 2026. Should i have expressed a different intent? counterfactual generation for LLM-based autonomous control. In *Forty-third International Conference on Machine Learning*.

Nils Feldhus and Laura Kopf. 2025. Interpreting language models through concept descriptions: A survey. In *Proceedings of the 8th BlackboxNLP Workshop: Analyzing and Interpreting Neural Networks for NLP*, pages 149–162, Suzhou, China. Association for Computational Linguistics.

Tao Feng, Yicheng Li, Chenglin Li, Hao Chen, Fei Yu, and Yin Zhang. 2024. Teaching small language models reasoning through counterfactual distillation. In *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing*, pages 5831–5842, Miami, Florida, USA. Association for Computational Linguistics.

Tao Feng, Lizhen Qu, Niket Tandon, and Gholamreza Haffari. 2025a. IRIS: An iterative and integrated framework for verifiable causal discovery in the absence of tabular data. In *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 9400–9428, Vienna, Austria. Association for Computational Linguistics.

Tao Feng, Lizhen Qu, Niket Tandon, Zhuang Li, Xiaoxi Kang, and Gholamreza Haffari. 2025b. On the reliability of large language models for causal discovery. In *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 9565–9590, Vienna, Austria. Association for Computational Linguistics.

Xuan Feng, Bo An, Tianlong Gu, Liang Chang, Fengrui Hao, Peipeng Yu, and Shuai Zhao. 2026. C2PO: Diagnosing and disentangling bias shortcuts in LLMs. In *Findings of the Association for Computational Linguistics: ACL 2026*, pages 24496–24515, San Diego, California, United States. Association for Computational Linguistics.

Yijun Feng. 2025. Counterfactual probing for hallucination detection and mitigation in large language models. *Preprint*, arXiv:2508.01862.

Elias Frantar and Dan Alistarh. 2023. SparseGPT: Massive language models can be accurately pruned in one-shot. In *Proceedings of the 40th International Conference on Machine Learning*, volume 202 of *Proceedings of Machine Learning Research*, pages 10323–10337. PMLR.

Dennis Frauen, Marie Brockschmidt, Konstantin Hess, Haorui Ma, Yuchen Ma, Abdurahman Maarouf, Maresa Schröder, Jonas Schweisthal, Yuxin Wang, Athiya Deviyani, Sonali Parbhoo, Rahul G. Krishnan, and Stefan Feuerriegel. 2026. Causal methods for LLM development and evaluation. *Preprint*, arXiv:2605.25998. Published in KDD 2026.

Jörg Frohberg and Frank Binder. 2022. CRASS: A novel data set and benchmark to test counterfactual reasoning of large language models. In *Proceedings of the Thirteenth Language Resources and Evaluation Conference*.

Jiarun Fu, Lizhong Ding, Qiuning Wei, Yuhan Guo, Yurong Cheng, and Junyu Zhang. 2026. Counterfactual planning for generalizable agents' actions. In *Proceedings of the Fortieth AAAI Conference on Artificial Intelligence*. AAAI Press.

Jinglong Gao, Xiao Ding, Bing Qin, and Ting Liu. 2023. Is ChatGPT a good causal reasoner? a comprehensive evaluation. In *Findings of the Association for Computational Linguistics: EMNLP 2023*, pages 11111–11126, Singapore. Association for Computational Linguistics.

Yair Ori Gat, Nitay Calderon, Amir Feder, Alexander Chapanin, Amit Sharma, and Roi Reichart. 2024. Faithful explanations of black-box NLP models using LLM-generated counterfactuals. In *The Twelfth International Conference on Learning Representations*.

Atticus Geiger, Duligur Ibeling, Amir Zur, Maheep Chaudhary, Sonakshi Chauhan, Jing Huang, Aryaman Arora, Zhengxuan Wu, Noah Goodman, Christopher Potts, and Thomas Icard. 2025. Causal abstraction: A theoretical foundation for mechanistic interpretability. *Journal of Machine Learning Research*, 26(83):1–64.

Atticus Geiger, Zhengxuan Wu, Christopher Potts, Thomas Icard, and Noah Goodman. 2024. Finding alignments between interpretable causal variables and distributed neural representations. In *Proceedings of the Third Conference on Causal Learning and Reasoning*, volume 236 of *Proceedings of Machine Learning Research*, pages 160–187. PMLR.

Gael Gendron, Bao Trung Nguyen, Alex Yuxuan Peng, Michael Witbrock, and Gillian Dobbie. 2024a. Can large language models learn independent causal mechanisms? In *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing*, Miami, Florida, USA. Association for Computational Linguistics.

Gael Gendron, Joze M. Rozanec, Michael Witbrock, and Gillian Dobbie. 2024b. Counterfactual causal inference in natural language with large language models. In *Causality and Large Models @NeurIPS 2024*.

John Gkountouras, Matthias Lindemann, Phillip Lippe, Efstratios Gavves, and Ivan Titov. 2025. Language agents meet causality – bridging LLMs and causal world models. In *The Thirteenth International Conference on Learning Representations*.

Javier González and Aditya V. Nori. 2024. Does reasoning emerge? examining the probabilities of causation in large language models. In *Advances in Neural Information Processing Systems*, volume 37, pages 117737–117761. Curran Associates, Inc.

Navita Goyal, Hal Daumé, Alexandre Drouin, and Dhanya Sridhar. 2025. Causal differentiating concepts: Interpreting lm behavior via causal representation learning. In *Advances in Neural Information Processing Systems*, volume 38, Main Conference, pages 66238–66266. Curran Associates, Inc.

Aaron Grattafiori, Abhimanyu Dubey, Abhinav Jauhri, Abhinav Pandey, Abhishek Kadian, Ahmad Al-Dahle, Aiesha Letman, Akhil Mathur, Alan Schelten, Alex Vaughan, Amy Yang, Angela Fan, Anirudh Goyal, Anthony Hartshorn, Aobo Yang, Archi Mitra, Archie Sravankumar, Artem Korenev, Arthur Hinsvark, and 542 others. 2024. The llama 3 herd of models. *Preprint*, arXiv:2407.21783.

Ziyin Gu, Jingyao Wang, Ran Zuo, Chuxiong Sun, Zeen Song, Changwen Zheng, and Wenwen Qiang. 2026. Group causal policy optimization for post-training large language models. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 40, pages 30834–30842.

Lin Guan, Karthik Valmeekam, Sarath Sreedharan, and Subbarao Kambhampati. 2023. Leveraging pre-trained large language models to construct and utilize world models for model-based task planning. In *Advances in Neural Information Processing Systems*, volume 36, pages 79081–79094. Curran Associates, Inc.

E.J. Gumbel. 1954. *Statistical Theory of Extreme Values and Some Practical Applications: A Series of Lectures*. Applied mathematics series. U.S. Government Printing Office.

Hanze Guo, Jing Yao, Xiao Zhou, Xiaoyuan Yi, and Xing Xie. 2025a. Counterfactual reasoning for steerable pluralistic value alignment of large language models. In *Advances in Neural Information Processing Systems*, volume 38, Main Conference, pages 122128–122169. Curran Associates, Inc.

Siyi Guo, Myrl G Marmarelis, Fred Morstatter, and Kristina Lerman. 2025b. Estimating causal effects of text interventions leveraging LLMs. In *Proceedings of the 14th International Joint Conference on Natural Language Processing and the 4th Conference of the Asia-Pacific Chapter of the Association for Computational Linguistics*, pages 170–190, Mumbai, India. The Asian Federation of Natural Language Processing and The Association for Computational Linguistics.

Wangzhen Guo, Qinkang Gong, Yanghui Rao, and Hanjiang Lai. 2023. Counterfactual multihop QA: A cause-effect approach for reducing disconnected reasoning. In *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 4214–4226, Toronto, Canada. Association for Computational Linguistics.

Yiju Guo, Wenkai Yang, Zexu Sun, Ning Ding, Zhiyuan Liu, and Yankai Lin. 2025c. Learning to focus: Causal attention distillation via gradient-guided token pruning. In *Advances in Neural Information Processing Systems*, volume 38, Main Conference, pages 24921–24948. Curran Associates, Inc.

Mukur Gupta, Noopur Bhatt, and Suman Jana. 2025a. CodeSCM: Causal analysis for multi-modal code generation. In *Proceedings of the 2025 Conference of the Nations of the Americas Chapter of the Association for Computational Linguistics: Human Language Technologies (Volume 1: Long Papers)*, pages 6779–6793, Albuquerque, New Mexico. Association for Computational Linguistics.

Mukur Gupta, Noopur Bhatt, and Suman Jana. 2025b. CodeSCM: Causal analysis for multi-modal code generation. In *Proceedings of the 2025 Conference of the Nations of the Americas Chapter of the Association for Computational Linguistics: Human Language Technologies (Volume 1: Long Papers)*, pages 6779–6793, Albuquerque, New Mexico. Association for Computational Linguistics.

Dongyoon Hahm, Woogyeol Jin, June Suk Choi, Sungsoo Ahn, and Kimin Lee. 2025. Enhancing LLM agent safety via causal influence prompting. In *Findings of the Association for Computational Linguistics: ACL 2025*, Vienna, Austria. Association for Computational Linguistics.

Kairong Han, Wenshuo Zhao, Ziyu Zhao, Ye Jun Jian, Lujia Pan, and Kun Kuang. 2025. CAT: Causal attention tuning for injecting fine-grained causal knowledge into large language models. In *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing*, pages 9904–9921, Suzhou, China. Association for Computational Linguistics.

Xiaochuang Han, Byron C. Wallace, and Yulia Tsvetkov. 2020. Explaining black box predictions and unveiling data artifacts through influence functions. In *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics*, pages 5553–5563, Online. Association for Computational Linguistics.

Yunseok Han, Yejoon Lee, and Jaeyoung Do. 2026. RFEval: Benchmarking reasoning faithfulness under counterfactual reasoning intervention in large reasoning models. In *The Fourteenth International Conference on Learning Representations*.

Shibo Hao, Yi Gu, Haodi Ma, Joshua Hong, Zhen Wang, Daisy Wang, and Zhiting Hu. 2023. Reasoning with language model is planning with world model. In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*, pages 8154–8173, Singapore. Association for Computational Linguistics.

Paul He, Yinya Huang, Mrinmaya Sachan, and Zhijing Jin. 2026a. Uncovering hidden correctness in LLM causal reasoning via symbolic verification. In *Proceedings of the 19th Conference of the European Chapter of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 1231–1250, Rabat, Morocco. Association for Computational Linguistics.

Yu He, Haozhe Zhu, Yiming Li, Shuo Shao, Hongwei Yao, Zhihao Liu, and Zhan Qin. 2026b. AttriGuard: Defeating indirect prompt injection in LLM agents via causal attribution of tool invocations. In *35th USENIX Security Symposium (USENIX Security 26)*, pages 1547–1566, Baltimore, MD. USENIX Association.

Jennifer L. Hill. 2011. Bayesian nonparametric modeling for causal inference. *Journal of Computational and Graphical Statistics*, 20(1):217–240.

Pingjun Hong and Benjamin Roth. 2026. Do llm self-explanations help users predict model behavior? evaluating counterfactual simulatability with pragmatic perturbations. *Preprint*, arXiv:2601.03775.

Ashish Hooda, Mihai Christodorescu, Miltiadis Allamanis, Aaron Wilson, Kassem Fawaz, and Somesh Jha. 2024. Do large code models understand programming concepts? counterfactual analysis for code predicates. In *Proceedings of the 41st International Conference on Machine Learning*, ICML'24. JMLR.org.

Zhizhang Hu, Yue Zhang, Ryan Rossi, Tong Yu, Sungchul Kim, and Shijia Pan. 2024. Are large language models capable of causal reasoning for sensing data analysis? In *Proceedings of the Workshop on Edge and Mobile Foundation Models*, EdgeFM '24, page 24–29, New York, NY, USA. Association for Computing Machinery.

Jing Huang, Zhengxuan Wu, Christopher Potts, Mor Geva, and Atticus Geiger. 2024a. RAVEL: Evaluating interpretability methods on disentangling language model representations. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 8669–8687, Bangkok, Thailand. Association for Computational Linguistics.

Xiaowei Huang, Wenjie Ruan, Wei Huang, Gaojie Jin, Yi Dong, Changshun Wu, Saddek Bensalem, Ronghui Mu, Yi Qi, Xingyu Zhao, Kaiwen Cai, Yanghao Zhang, Sihao Wu, Peipei Xu, Dengyu Wu, Andre Freitas, and Mustafa A. Mustafa. 2024b. A survey of safety and trustworthiness of large language models through the lens of verification and validation. *Artificial Intelligence Review*, 57(7):175.

Yinya Huang, Ruixin Hong, Hongming Zhang, Wei Shao, Zhicheng Yang, Dong Yu, Changshui Zhang, Xiaodan Liang, and Linqi Song. 2024c. CLOMO: Counterfactual logical modification with large language models. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 11012–11034, Bangkok, Thailand. Association for Computational Linguistics.

Yinya Huang, Ruixin Hong, Hongming Zhang, Wei Shao, Zhicheng Yang, Dong Yu, Changshui Zhang, Xiaodan Liang, and Linqi Song. 2024d. CLOMO: Counterfactual logical modification with large language models. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 11012–11034, Bangkok, Thailand. Association for Computational Linguistics.

Alihan Hüyük, Xinnuo Xu, Jacqueline Maasch, Aditya Nori, and Javier Hernandez. 2025. Reasoning elicitation in language models via counterfactual feedback. In *International Conference on Learning Representations*, volume 2025, pages 76804–76832.

Ujun Jeong, Saketh Vishnubhatla, Bohan Jiang, Andre Harrison, Adrienne Raglin, and Huan Liu. 2026. Large language models for causal relations extraction in social media: A validation framework for disaster intelligence. *Preprint*, arXiv:2605.11348.

Zhenlan Ji, Pingchuan Ma, Zongjie Li, Zhaoyu Wang, and Shuai Wang. 2025. Causality-aided evaluation and explanation of large language model-based code generation. *Association for Computing Machinery*, 2(ISSTA).

Bohan Jiang, Pingchuan Ma, Zhuoyu Shi, Fred Morstatter, Adrienne Raglin, and Huan Liu. 2026. On causal and anticausal llm-based data synthesis. In *Proceedings of the Nineteenth ACM International Conference on Web Search and Data Mining*, WSDM '26, page 1160–1164, New York, NY, USA. Association for Computing Machinery.

Haitao Jiang, Lin Ge, Yuhe Gao, Jianian Wang, and Rui Song. 2024. LLM4causal: Democratized causal tools for everyone via large language model. In *First Conference on Language Modeling*.

Zhengbao Jiang, Jun Araki, Haibo Ding, and Graham Neubig. 2022. Understanding and improving zero-shot multi-hop reasoning in generative question answering. In *Proceedings of the 29th International Conference on Computational Linguistics*, pages 1765–1775, Gyeongju, Republic of Korea. International Committee on Computational Linguistics.

Youngjin Jin, Hanna Kim, Kwanwoo Kim, Chanhee Lee, and Seungwon Shin. 2026. Argora: Orchestrated argumentation for causally grounded llm reasoning and decision making. *Preprint*, arXiv:2601.21533.

Zhijing Jin, Yuen Chen, Felix Leeb, Luigi Gresele, Ojasv Kamal, Zhiheng LYU, Kevin Blin, Fernando Gonzalez Adauto, Max Kleiman-Weiner, Mrinmaya Sachan, and Bernhard Schölkopf. 2023. CLadder: A benchmark to assess causal reasoning capabilities of language models. In *Thirty-seventh Conference on Neural Information Processing Systems*.

Zhijing Jin, Jiarui Liu, Zhiheng LYU, Spencer Poff, Mrinmaya Sachan, Rada Mihalcea, Mona T. Diab, and Bernhard Schölkopf. 2024. Can large language models infer causation from correlation? In *The Twelfth International Conference on Learning Representations*.

Thomas Jiralerspong, Xiaoyin Chen, Yash More, Vedant Shah, and Yoshua Bengio. 2024. Efficient causal graph discovery using large language models. In *ICLR 2024 Workshops: AGI*.

Mohit Jiwatode, Alexander Dockhorn, and Bodo Rosenhahn. 2026. From gameplay traces to game mechanics: Causal induction with large language models. *Preprint*, arXiv:2602.00190.

Anish R Joishy, Ishwar B Balappanawar, Vamshi Krishna Bonagiri, Manas Gaur, Krishnaprasad Thirunarayan, and Ponnurangam Kumaraguru. 2026. Flying pigs, far and beyond: Evaluating llm reasoning in counterfactual worlds. *Preprint*, arXiv:2505.22318.

Abhinav Joshi, Areeb Ahmad, and Ashutosh Modi. 2024a. Cold: Causal reasoning in closed daily activities. In *Advances in Neural Information Processing Systems*, volume 37, pages 5145–5187. Curran Associates, Inc.

Nitish Joshi, Abulhair Saparov, Yixin Wang, and He He. 2024b. LLMs are prone to fallacies in causal inference. In *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing*, pages 10553–10569, Miami, Florida, USA. Association for Computational Linguistics.

Jean Kaddour, Aengus Lynch, Qi Liu, Matt J Kusner, and Silva Ricardo. 2025. Causal machine learning: A survey and open problems. *Foundations and Trends in Optimization*, 9(1-2):1–247.

Kacper Kadziolka and Saber Salehkaleybar. 2025. Causal reasoning in pieces: Modular in-context learning for causal discovery. *arXiv preprint arXiv:2507.23488*.

Shiv Kampani, David Hidary, Constantijn van der Poel, Martin Ganahl, and Brenda Miao. 2024. LLM-initialized differentiable causal discovery. In *Causality and Large Models @NeurIPS 2024*.

Adam Karvonen, Euan Ong, Subhash Kantamneni, and Samuel Marks. 2026. Would this change your answer? evaluating explanations of llm behavior in the wild with counterfactual experiments. *Preprint*, arXiv:2608.16747.

Tejas Kasetty, Divyat Mahajan, Gintare Karolina Dziugaite, Alexandre Drouin, and Dhanya Sridhar. 2024. Evaluating interventional reasoning capabilities of large language models. In *Causality and Large Models @NeurIPS 2024*.

Elahe Khatibi, Mahyar Abbasian, Zhongqi Yang, Iman Azimi, and Amir M. Rahmani. 2025. Alcm: Autonomous llm-augmented causal discovery framework. *Preprint*, arXiv:2405.01744.

Emre Kiciman, Robert Ness, Amit Sharma, and Chenhao Tan. 2024. Causal reasoning and large language models: Opening a new frontier for causality. *Transactions on Machine Learning Research*. Featured Certification.

Hyunjun Kim. 2026. Holograph: Active causal discovery via sheaf-theoretic alignment of large language model priors. *Preprint*, arXiv:2512.24478.

Jongho Kim and Seung-won Hwang. 2025. Counterfactual-consistency prompting for relative temporal understanding in large language models. In *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 2: Short Papers)*, pages 1210–1225, Vienna, Austria. Association for Computational Linguistics.

Yejin Kim, Eojin Kang, Juae Kim, and H. Howie Huang. 2024. Causal reasoning in large language models: A knowledge graph approach. In *NeurIPS 2024 Workshop on Causality and Large Models*.

Kasia Kobalczyk and Mihaela Van Der Schaar. 2025. Preference learning for AI alignment: a causal perspective. In *Proceedings of the 42nd International Conference on Machine Learning*, volume 267 of *Proceedings of Machine Learning Research*, pages 31063–31083. PMLR.

Peter Kochelka, Aleš Manuel Papáček, Vojtěch Dvořák, and Ondřej Dušek. 2026. Do llms make more mistakes if they do not believe the input data? *Preprint*, arXiv:2609.09363.

Amit Kumar, Elnur Adl Zarabi, Suranjana Trivedy, Zhiqian Chen, Lei Zhang, Kaiqun Fu, and Taoran Ji. 2026. From causal plausibility to causal reliability: Evaluating LLMs as calibrated direct causal-edge classifiers. *arXiv preprint arXiv:2608.23660*.

Emre Kuru and Reyhan Aydoğan. 2026. Effective use of llms via counterfactual reasoning for transparent recommendation explanations. In *PRIMA 2025: Principles and Practice of Multi-Agent Systems*, pages 277–285, Cham. Springer Nature Switzerland.

Brenden M. Lake, Tomer D. Ullman, Joshua B. Tenenbaum, and Samuel J. Gershman. 2017. Building machines that learn and think like people. *Behavioral and Brain Sciences*, 40:e253.

Su Lan, Xuefei Yin, Yanming Zhu, and Alan Wee-Chung Liew. 2026. Causal-audit: Explicit and auditable graph-based reasoning via target-aware causal chain construction. In *Findings of the Association for Computational Linguistics: ACL 2026*, pages 12484–12500, San Diego, California, United States. Association for Computational Linguistics.

Andrej Leban and Yuekai Sun. 2026. Causalds: Benchmarking causal reasoning in data-science agents. *Preprint*, arXiv:2607.08093.

Chanhui Lee, Juhyeon Kim, YongJun Jeong, Yoonseok Yeom, Juhyun Lyu, Jung-Hee Kim, Sangmin Lee, Sangjun Han, Hyeokjun Choe, Soyeon Park, Woohyung Lim, Kyunghoon Bae, Sungbin Lim, and Sanghack Lee. 2024. On incorporating prior knowledge extracted from pre-trained language models into causal discovery. In *Causality and Large Models @NeurIPS 2024*.

Po-Han Lee, Yu-Cheng Lin, Chan-Tung Ku, Chan Hsu, Pei-Cing Huang, Ping-Hsun Wu, and Yihuang Kang. 2025. LLM-Based Agents for Automated Confounder Discovery and Subgroup Analysis in Causal Inference . In *2025 IEEE International Conference on Information Reuse and Integration and Data Science (IRI)*, pages 265–270, Los Alamitos, CA, USA. IEEE Computer Society.

Patrick Lewis, Ethan Perez, Aleksandra Piktus, Fabio Petroni, Vladimir Karpukhin, Naman Goyal, Heinrich Küttler, Mike Lewis, Wen-tau Yih, Tim Rocktäschel, Sebastian Riedel, and Douwe Kiela. 2020. Retrieval-augmented generation for knowledge-intensive nlp tasks. In *Advances in Neural Information Processing Systems*, volume 33, pages 9459–9474. Curran Associates, Inc.

Bowen Li, Ziqi Xu, Jing Ren, Renqiang Luo, Xikun Zhang, Xiuzhen Zhang, Yongli Ren, and Feng Xia. 2026a. Debiasing large language models via adaptive causal prompting with sketch-of-thought. In *Findings of the Association for Computational Linguistics: EACL 2026*, pages 4481–4499, Rabat, Morocco. Association for Computational Linguistics.

Jiang Li, Zehua Duo, Guanglai Gao, and Xiangdong Su. 2026b. CausalityCheck: A framework for evaluating causal reasoning in large language models. In *Findings of the Association for Computational Linguistics: ACL 2026*, pages 16403–16429, San Diego, California, United States. Association for Computational Linguistics.

Jiaxuan Li, Lang Yu, and Allyson Ettinger. 2023. Counterfactual reasoning: Testing language models' understanding of hypothetical scenarios. In *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (Volume 2: Short Papers)*, pages 804–815, Toronto, Canada. Association for Computational Linguistics.

Jingling Li, Zeyu Tang, Xiaoyu Liu, Peter Spirtes, Kun Zhang, Liu Leqi, and Yang Liu. 2025a. Prompting fairness: Integrating causality to debias large language models. In *The Thirteenth International Conference on Learning Representations*.

Junyi Li, Yongqiang Chen, and Ningning Ding. 2026c. CiPO: Counterfactual unlearning for large reasoning models through iterative preference optimization. In *Proceedings of the 64th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 3152–3170, San Diego, California, United States. Association for Computational Linguistics.

Junyi Li, Yongqiang Chen, Chenxi Liu, Qianyi Cai, Tongliang Liu, Bo Han, Kun Zhang, and Hui Xiong. 2025b. Can large language models help experimental design for causal discovery? In *ICML 2025 Workshop on Scaling Up Intervention Models*.

Mengfan Li, Xuanhua Shi, and Yang Deng. 2026d. CoSToM: Causal-oriented steering for intrinsic theory-of-mind alignment in large language models. In *Proceedings of the 64th Annual Meeting of the Association*

for Computational Linguistics (Volume 1: Long Papers), pages 9302–9317, San Diego, California, United States. Association for Computational Linguistics.

Peiwen Li, Xin Wang, Zeyang Zhang, Yuan Meng, Fang Shen, Yue Li, Jialong Wang, Yang Li, and Wenwu Zhu. 2024a. RealTCD: Temporal causal discovery from interventional data with large language model. In *Proceedings of the 33rd ACM International Conference on Information and Knowledge Management*, pages 4669–4677. Association for Computing Machinery.

Pengfeng Li, Chen Huang, Chaoqun Hao, Hongyao Chen, Xiao-Yong Wei, Wenqiang Lei, and See-Kiong Ng. 2026e. METER: Evaluating multi-level contextual causal reasoning in large language models. In *Proceedings of the 64th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 36049–36073, San Diego, California, United States. Association for Computational Linguistics.

Shuaike Li, Kai Zhang, Xianquan Wang, Jiachen Liu, and Shengpeng Mo. 2026f. From fact overwriting to knowledge evolution: Causal editing via on-policy self-distillation. *Preprint*, arXiv:2605.28303.

Xin Li, Zhuo Cai, Shoujin Wang, Kun Yu, and Fang Chen. 2025c. A survey on enhancing causal reasoning ability of large language models. *Preprint*, arXiv:2503.09326. Version 1.

Yongqi Li, Mayi Xu, Xin Miao, Shen Zhou, and Tieyun Qian. 2024b. Prompting large language models for counterfactual generation: An empirical study. In *Proceedings of the 2024 Joint International Conference on Computational Linguistics, Language Resources and Evaluation (LREC-COLING 2024)*, pages 13201–13221, Torino, Italia. ELRA and ICCL.

Yuangang Li, Yiqing Shen, Yi Nian, Jiechao Gao, Ziyi Wang, Chenxiao Yu, Li Li, Jie Wang, Xiyang Hu, and Yue Zhao. 2026g. Mitigating hallucinations in large language models via causal reasoning. In *Proceedings of the AAAI Conference on Artificial Intelligence*.

Zhiyuan Li, Linyuan Gao, Xuechun Ding, Hongwei Chen, Yuan Wu, and Yi Chang. 2026h. Caskg: Counterfactual-causal skill graphs for scalable agent skill retrieval. *Preprint*, arXiv:2608.25500.

Zihao Li and Fabrizio Russo. 2026. Leveraging large language models for causal discovery: a constraint-based, argumentation-driven approach. In *Proceedings of the 42nd Conference on Uncertainty in Artificial Intelligence*, volume 337 of *Proceedings of Machine Learning Research*, pages 3631–3667. PMLR.

Marvin Limpijankit, Yanda Chen, Melanie Subbiah, Nicholas Deas, and Kathleen McKeown. 2025. Counterfactual simulatability of LLM explanations for generation tasks. In *Proceedings of the 18th International Natural Language Generation Conference*, pages 659–683, Hanoi, Vietnam. Association for Computational Linguistics.

Bo Liu, Hongyan Li, and Shenda Hong. 2026. Llm-gc: Advancing granger causal discovery from time series with multimodel language modeling. In *Proceedings of the Nineteenth ACM International Conference on Web Search and Data Mining*, WSDM '26, page 387–395, New York, NY, USA. Association for Computing Machinery.

Chenxi Liu, Yongqiang Chen, Tongliang Liu, Mingming Gong, James Cheng, Bo Han, and Kun Zhang. 2024a. Discovery of the hidden world with large language models. In *Advances in Neural Information Processing Systems*, volume 37, pages 102307–102365. Curran Associates, Inc.

Lijia Liu, Takumi Kondo, Kyohei Atarashi, Koh Takeuchi, Jiyi Li, Shigeru Saito, and Hisashi Kashima. 2025a. Counterfactual evaluation for blind attack detection in LLM-based evaluation systems. In *Proceedings of the 14th International Joint Conference on Natural Language Processing and the 4th Conference of*

the Asia-Pacific Chapter of the Association for Computational Linguistics, Mumbai, India. The Asian Federation of Natural Language Processing and The Association for Computational Linguistics.

Shuxin Liu and Jianhao Zhang. 2026. CAKE: Causal-guided adaptive knowledge editing for LLMs. In *Proceedings of the 64th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 20040–20070, San Diego, California, United States. Association for Computational Linguistics.

Xiao Liu, Zirui Wu, Xueqing Wu, Pan Lu, Kai-Wei Chang, and Yansong Feng. 2024b. Are LLMs capable of data-based statistical and causal reasoning? benchmarking advanced quantitative reasoning with data. In *Findings of the Association for Computational Linguistics: ACL 2024*, pages 9215–9235, Bangkok, Thailand. Association for Computational Linguistics.

Xiao Liu, Da Yin, Chen Zhang, Yansong Feng, and Dongyan Zhao. 2023a. The magic of IF: Investigating causal reasoning abilities in large language models of code. In *Findings of the Association for Computational Linguistics: ACL 2023*, pages 9009–9022, Toronto, Canada. Association for Computational Linguistics.

Xiao Liu, Da Yin, Chen Zhang, Dongyan Zhao, and Yansong Feng. 2025b. Eliciting and improving the causal reasoning abilities of large language models with conditional statements. *Computational Linguistics*, 51:467–504.

Xiaoyu Liu, Paiheng Xu, Junda Wu, Jiaxin Yuan, Yifan Yang, Yuhang Zhou, Fuxiao Liu, Tianrui Guan, Haoliang Wang, Tong Yu, Julian McAuley, Wei Ai, and Furong Huang. 2025c. Large language models and causal inference in collaboration: A comprehensive survey. In *Findings of the Association for Computational Linguistics: NAACL 2025*, pages 7683–7699, Albuquerque, New Mexico. Association for Computational Linguistics.

Yang Liu, Yuanshun Yao, Jean-Francois Ton, Xiaoying Zhang, Ruocheng Guo, Hao Cheng, Yegor Klochkov, Muhammad Faaiz Taufiq, and Hang Li. 2023b. Trustworthy LLMs: A Survey and Guideline for Evaluating Large Language Models' Alignment. In *NeurIPS 2023 Workshop on Socially Responsible Language Modelling Research (SoLaR)*.

Yi Liu, Lianzhe Huang, Shicheng Li, Sishuo Chen, Hao Zhou, Fandong Meng, Jie Zhou, and Xu Sun. 2023c. Recall: A benchmark for llms robustness against external counterfactual knowledge. *Preprint*, arXiv:2311.08147.

Yi Liu, Lianzhe Huang, Shicheng Li, Sishuo Chen, Hao Zhou, Fandong Meng, Jie Zhou, and Xu Sun. 2023d. Recall: A benchmark for llms robustness against external counterfactual knowledge. *Preprint*, arXiv:2311.08147.

Yiran Liu, Xiaoang Xu, Zhiyi Hou, and Yang Yu. 2024c. Causality based front-door defense against backdoor attack on language models. In *Forty-first International Conference on Machine Learning*.

Yujian Liu, Yang Zhang, Tommi Jaakkola, and Shiyu Chang. 2024d. Revisiting who's harry potter: Towards targeted unlearning from a causal intervention perspective. In *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing*, pages 8708–8731, Miami, Florida, USA. Association for Computational Linguistics.

Yutian Liu, Zhengyi Yang, Jiancan Wu, and Xiang Wang. 2025d. Enhancing temporal sensitivity of large language model for recommendation with counterfactual tuning. *Preprint*, arXiv:2507.03047.

Zheyuan Liu, Guangyao Dou, Zhaoxuan Tan, Yijun Tian, and Meng Jiang. 2024e. Towards safer large language models through machine unlearning. In *Findings of the Association for Computational Linguistics: ACL 2024*, pages 1817–1829, Bangkok, Thailand. Association for Computational Linguistics.

Zhongwen Luan, Xiaoyu Zhang, Ming Hu, Yue Yang, Jiongchi Yu, and Xiaohong Chen. 2026. Repair or resample? rethinking failure debugging in LLM multi-agent systems. *Preprint*, arXiv:2608.25920.

Zhiheng Lyu, Zhijing Jin, Fernando Gonzalez Adauto, Rada Mihalcea, Bernhard Schölkopf, and Mrinmaya Sachan. 2024. Do LLMs think fast and slow? a causal study on sentiment analysis. In *Findings of the Association for Computational Linguistics: EMNLP 2024*, pages 9353–9372, Miami, Florida, USA. Association for Computational Linguistics.

Zongyi Lyu, Zhenlan Ji, Songqiang Chen, Liwen Wang, Yuheng Huang, Shuai Wang, and Shing-Chi Cheung. 2026. Cam: A causality-based analysis framework for multi-agent code generation systems. *Proceedings of the ACM on Software Engineering*, 3(ISSTA):3364–3388.

Jiatong Ma, Linmei Hu, Rang Li, and Wenbo Fu. 2025. Local: Logical and causal fact-checking with llm-based multi-agents. In *Proceedings of the ACM on Web Conference 2025*, WWW '25, page 1614–1625, New York, NY, USA. Association for Computing Machinery.

Jing Ma. 2025. Causal inference with large language model: A survey. In *Findings of the Association for Computational Linguistics: NAACL 2025*, pages 5901–5913, Albuquerque, New Mexico. Association for Computational Linguistics.

Jacqueline R. M. A. Maasch, Alihan Hüyük, Xinnuo Xu, Aditya V. Nori, and Javier Gonzalez. 2025. Compositional causal reasoning evaluation in language models. In *Forty-second International Conference on Machine Learning*.

Rahul Madhavan, Rishabh Garg, Kahini Wadhawan, and Sameep Mehta. 2023. CFL: Causally fair language models through token-level attribute controlled generation. In *Findings of the Association for Computational Linguistics: ACL 2023*, pages 11344–11358, Toronto, Canada. Association for Computational Linguistics.

Andreas Madsen, Sarath Chandar, and Siva Reddy. 2024. Are self-explanations from large language models faithful? In *Findings of the Association for Computational Linguistics: ACL 2024*, pages 295–337, Bangkok, Thailand. Association for Computational Linguistics.

Giacomo Magnifico. 2025. Automated classification of causal relations. evaluating different LLM performances. In *Proceedings of the 9th Student Research Workshop associated with the International Conference Recent Advances in Natural Language Processing*, pages 27–36, Varna, Bulgaria. INCOMA Ltd., Shoumen, Bulgaria.

Katie Matton, Robert Ness, John Guttag, and Emre Kiciman. 2025. Walk the talk? measuring the faithfulness of large language model explanations. In *The Thirteenth International Conference on Learning Representations*.

Fabrizio Maturo, Donato Riccio, Andrea Mazzitelli, Giuseppe Bifulco, Francesco Paolone, and Iulia Brezeanu. 2025. Arcadia: Scalable causal discovery for corporate bankruptcy analysis using agentic ai. *Preprint*, arXiv:2512.00839.

Kawin Mayilvaghanan, Siddhant Gupta, and Ayush Kumar. 2026. Counterfactual fairness evaluation of LLM-based contact center agent quality assurance system. In *Findings of the Association for Computational Linguistics: ACL 2026*, pages 37914–37952, San Diego, California, United States. Association for Computational Linguistics.

Harry Mayne, Ryan Othniel Kearns, Yushi Yang, Andrew M. Bean, Eoin D. Delaney, Chris Russell, and Adam Mahdi. 2025. LLMs don't know their own decision boundaries: The unreliability of self-generated counterfactual explanations. In *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing*, pages 24161–24186, Suzhou, China. Association for Computational Linguistics.

Joshua Maynez, Shashi Narayan, Bernd Bohnet, and Ryan McDonald. 2020. On faithfulness and factuality in abstractive summarization. In *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics*, pages 1906–1919, Online. Association for Computational Linguistics.

Kevin Meng, David Bau, Alex Andonian, and Yonatan Belinkov. 2022. Locating and editing factual associations in gpt. In *Proceedings of the 36th International Conference on Neural Information Processing Systems*, NIPS '22, Red Hook, NY, USA. Curran Associates Inc.

X. Miao, Y. Li, H. Kang, and 1 others. 2026. How robust are large language models against word-level spurious correlations? a causal discovery approach. *Machine Learning*, 115:59.

Xin Miao, Yongqi Li, Shen Zhou, and Tieyun Qian. 2024. Episodic memory retrieval from LLMs: A neuromorphic mechanism to generate commonsense counterfactuals for relation extraction. In *Findings of the Association for Computational Linguistics: ACL 2024*, pages 2489–2511, Bangkok, Thailand. Association for Computational Linguistics.

Martina Miliani, Serena Auriemma, Alessandro Bondielli, Emmanuele Chersoni, Lucia C. Passaro, Irene Sucameli, and Alessandro Lenci. 2025. ExpliCa: Evaluating explicit causal reasoning in large language models. In *Findings of the Association for Computational Linguistics: ACL 2025*, pages 17335–17355, Vienna, Austria. Association for Computational Linguistics.

Moritz Miller, Bernhard Schölkopf, and Siyuan Guo. 2025. Counterfactual reasoning: an analysis of in-context emergence. In *The Thirty-ninth Annual Conference on Neural Information Processing Systems*.

Avni Mittal and Rauno Arike. 2026. C2-faith: Benchmarking LLM judges for causal and coverage faithfulness in chain-of-thought reasoning. In *Proceedings of the Fifth Workshop on Generation, Evaluation and Metrics (GEM)*, pages 678–691, San Diego, California, USA. Association for Computational Linguistics.

Kaijie Mo, Siddhartha Venkatayogi, Chantal Shaib, Ramez Kouzy, Wei Xu, Byron C Wallace, and Junyi Jessy Li. 2026. Faithfulness vs. safety: Evaluating LLM behavior under counterfactual medical evidence. In *Findings of the Association for Computational Linguistics: ACL 2026*, pages 37053–37081, San Diego, California, United States. Association for Computational Linguistics.

Joris M Mooij, Jonas Peters, Dominik Janzing, Jakob Zscheischler, and Bernhard Schölkopf. 2016. Distinguishing cause from effect using observational data: methods and benchmarks. *Journal of Machine Learning Research*, 17(32):1–102.

Ramaravind K. Mothilal, Amit Sharma, and Chenhao Tan. 2020. Explaining machine learning classifiers through diverse counterfactual explanations. In *Proceedings of the 2020 Conference on Fairness, Accountability, and Transparency*, pages 607–617.

Aaron Mueller, Atticus Geiger, Sarah Wiegreffe, Dana Arad, Iván Arcuschin, Adam Belfki, Yik Siu Chan, Jaden Fried Fiotto-Kaufman, Tal Haklay, Michael Hanna, Jing Huang, Rohan Gupta, Yaniv Nikankin, Hadas Orgad, Nikhil Prakash, Anja Reusch, Aruna Sankaranarayanan, Shun Shao, Alessandro Stolfo, and 4 others. 2025. MIB: A mechanistic interpretability benchmark. In *Proceedings of the 42nd International Conference on Machine Learning*, volume 267 of *Proceedings of Machine Learning Research*, pages 45069–45108. PMLR.

Thushara Manjari Naduvilakandy, Hyeju Jang, and Mohammad Al Hasan. 2025. Retrieval augmented generation based large language models for causality mining. *Preprint*, arXiv:2505.23944.

Yash Narendra and Niladri Chatterjee. 2026. Causal what-ifs: Rethinking counterfactuals with llm agents. In *Neural Information Processing*, pages 184–198, Singapore. Springer Nature Singapore.

Minh Hoang Nguyen, Van Dai Do, Dung Nguyen, Thin Nguyen, and Hung Le. 2025. Causalplan: Empowering efficient llm multi-agent collaboration through causality-driven planning. *Preprint*, arXiv:2508.13721.

Van Bach Nguyen, Christin Seifert, and Jörg Schlötterer. 2024. CEval: A benchmark for evaluating counterfactual text generation. In *Proceedings of the 17th International Natural Language Generation Conference*, pages 55–69, Tokyo, Japan. Association for Computational Linguistics.

Van Bach Nguyen, Christin Seifert, and Jörg Schlötterer. 2026a. Guiding llms to generate high-fidelity and high-quality counterfactual explanations for text classification. In *Explainable Artificial Intelligence*, pages 158–176, Cham. Springer Nature Switzerland.

Vy Nguyen, Ziqi Xu, Jeffrey Chan, Estrid He, Feng Xia, and Xiuzhen Zhang. 2026b. Hallucinate less by thinking more: aspect-based causal abstention for large language models. In *Proceedings of the Fortieth AAAI Conference on Artificial Intelligence and Thirty-Eighth Conference on Innovative Applications of Artificial Intelligence and Sixteenth Symposium on Educational Advances in Artificial Intelligence*, AAAI'26/IAAI'26/EAAI'26. AAAI Press.

Nirit Nussbaum-Hoffer, Nitay Calderon, Liat Ein-Dor, and Roi Reichart. 2026. Llm explainability with counterfactual chains and causal graphs. *Preprint*, arXiv:2606.05972.

Keane Ong, Rui Mao, Deeksha Varshney, Paul Pu Liang, Erik Cambria, and Gianmarco Mengaldo. 2025. Deriving strategic market insights with large language models: A benchmark for forward counterfactual generation. In *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing*, pages 11411–11434, Suzhou, China. Association for Computational Linguistics.

Francesco Ortu, Zhijing Jin, Diego Doimo, Mrinmaya Sachan, Alberto Cazzaniga, and Bernhard Schölkopf. 2024. Competition of mechanisms: Tracing how language models handle facts and counterfactuals. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 8420–8436, Bangkok, Thailand. Association for Computational Linguistics.

Leyi Ouyang. 2025. Interpreting public sentiment in diplomacy events: A counterfactual analysis framework using large language models. *Preprint*, arXiv:2509.20367.

Long Ouyang, Jeffrey Wu, Xu Jiang, Diogo Almeida, Carroll Wainwright, Pamela Mishkin, Chong Zhang, Sandhini Agarwal, Katarina Slama, Alex Ray, John Schulman, Jacob Hilton, Fraser Kelton, Luke Miller, Maddie Simens, Amanda Askell, Peter Welinder, Paul F Christiano, Jan Leike, and Ryan Lowe. 2022. Training language models to follow instructions with human feedback. In *Advances in Neural Information Processing Systems*, volume 35, pages 27730–27744. Curran Associates, Inc.

Haolin Pan, Lianghong Huang, Dong Jinyuan, Mingjie Xing, and Yanjun Wu. 2026. ECCO: Evidence-driven causal reasoning for compiler optimization. In *Forty-third International Conference on Machine Learning*.

Letitia Parcalabescu and Anette Frank. 2024. On measuring faithfulness or self-consistency of natural language explanations. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 6048–6089, Bangkok, Thailand. Association for Computational Linguistics.

Joon Sung Park, Joseph O'Brien, Carrie Jun Cai, Meredith Ringel Morris, Percy Liang, and Michael S. Bernstein. 2023. Generative agents: Interactive simulacra of human behavior. In *Proceedings of the 36th Annual ACM Symposium on User Interface Software and Technology*, UIST '23, New York, NY, USA. Association for Computing Machinery.

Alicia Parrish, Angelica Chen, Nikita Nangia, Vishakh Padmakumar, Jason Phang, Jana Thompson, Phu Mon Htut, and Samuel R. Bowman. 2022. BBQ: A hand-built bias benchmark for question answering. In *Findings of the Association for Computational Linguistics: ACL 2022*, pages 2086–2105, Dublin, Ireland. Association for Computational Linguistics.

Alessandra Parziale, Gianmario Voria, Valeria Pontillo, Gemma Catolino, Andrea De Lucia, and Fabio Palomba. 2026a. SCOPE: A dataset of stereotyped prompts for counterfactual fairness assessment of LLMs. In *33rd IEEE International Conference on Software Analysis, Evolution and Reengineering (SANER 2026), Workshops and Tutorials Track*.

Alessandra Parziale, Gianmario Voria, Valeria Pontillo, Gemma Catolino, Andrea De Lucia, and Fabio Palomba. 2026b. Toward systematic counterfactual fairness evaluation of large language models: The caffe framework. In *Proceedings of the 2026 IEEE/ACM 48th International Conference on Software Engineering*, ICSE '26, page 3117–3129, New York, NY, USA. Association for Computing Machinery.

Judea Pearl. 2002. Reasoning with cause and effect. *AI Mag.*, 23(1):95–111.

Judea Pearl. 2009. *Causality: Models, Reasoning and Inference*, 2nd edition. Cambridge University Press, USA.

Bo Peng, Sirui Chen, Lei Xu, and Chaochao Lu. 2026. Causcientist: Teaching llms to respect data for causal discovery. *Preprint*, arXiv:2601.13614.

Jonas Peters and Peter Bühlmann. 2015. Structural intervention distance for evaluating causal graphs. *Neural Computation*, 27(3):771–799.

Fernando Cardenas Piepereit. 2026. Architecture-dependent causal transfer of activation states across large language models. *Preprint*, arXiv:2608.16347.

Edoardo Pona, Milad Kazemi, Yali Du, David Watson, and Nicola Paoletti. 2025. Abstract counterfactuals for language model agents. In *The Thirty-ninth Annual Conference on Neural Information Processing Systems*.

Shauli Ravfogel, Anej Svete, Vésteinn Snæbjarnarson, and Ryan Cotterell. 2025. Gumbel counterfactual generation from language models. In *The Thirteenth International Conference on Learning Representations*.

Jing Ren, Wenhao Zhou, Bowen Li, Mujie Liu, Nguyen Linh Dan Le, Jiade Cen, Liping Chen, Ziqi Xu, Xiwei Xu, and Xiaodong Li. 2025. Causal prompting for implicit sentiment analysis with large language models. *Preprint*, arXiv:2507.00389.

Melissa Roemmele, Cosmin Adrian Bejan, and Andrew S. Gordon. 2011. Choice of plausible alternatives: An evaluation of commonsense causal reasoning. In *AAAI Spring Symposium: Logical Formalizations of Commonsense Reasoning*.

Angelika Romanou, Syrielle Montariol, Debjit Paul, Leo Laugier, Karl Aberer, and Antoine Bosselut. 2023. CRAB: Assessing the strength of causal relationships between real-world events. In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*, pages 15198–15216, Singapore. Association for Computational Linguistics.

Donald B. Rubin. 1974. Estimating causal effects of treatments in randomized and nonrandomized studies. *Journal of Educational Psychology*, 66(5):688–701.

Karen Sachs, Omar Perez, Dana Pe'er, Douglas A. Lauffenburger, and Garry P. Nolan. 2005. Causal protein-signaling networks derived from multiparameter single-cell data. *Science*, 308(5721):523–529.

Soumadeep Saha, Akshay Chaturvedi, Saptarshi Saha, Utpal Garain, and Nicholas Asher. 2026. KisMATH: Do LLMs have knowledge of implicit structures in mathematical reasoning? *Transactions of the Association for Computational Linguistics*, 14:1308–1328.

Ryan Saklad, Aman Chadha, Oleg V. Pavlov, and Raha Moraffah. 2026. Can large language models infer causal relationships from real-world text? In *Proceedings of the 64th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 21957–21989, San Diego, California, United States. Association for Computational Linguistics.

Marc Saouda, Rajprakash Bale, Eren Aldis, and Cloves Almeida. 2026. Structure over scale: Schema-constrained causal graphs for rag. *Preprint*, arXiv:2607.22592.

Ayush Sawarni, Jiyuan Tan, and Vasilis Syrgkanis. 2026. Causalreasoningbenchmark: A real-world benchmark for disentangled evaluation of causal identification and estimation. In *RLEval: Methods and Reinforcement Learning Environments for Evaluating AI Agents*.

Bernhard Schölkopf, Francesco Locatello, Stefan Bauer, Nan Rosemary Ke, Nal Kalchbrenner, Anirudh Goyal, and Yoshua Bengio. 2021. Toward causal representation learning. *Proceedings of the IEEE*, 109(5):612–634.

Marco Scutari. 2010a. Learning bayesian networks with the bnlearn r package. *Journal of Statistical Software*, 35(3):1–22.

Marco Scutari. 2010b. Learning bayesian networks with the bnlearn r package. *Journal of statistical software*, 35(1):1–22.

Jaineet Shah. 2026. Causal agent replay: Counterfactual attribution for llm-agent failures. *Preprint*, arXiv:2606.08275.

Uri Shalit, Fredrik D. Johansson, and David Sontag. 2017. Estimating individual treatment effect: Generalization bounds and algorithms. In *Proceedings of the 34th International Conference on Machine Learning*.

Zhihong Shao, Peiyi Wang, Qihao Zhu, Runxin Xu, Junxiao Song, Mingchuan Zhang, Y. K. Li, Y. Wu, and Daya Guo. 2024. Deepseekmath: Pushing the limits of mathematical reasoning in open language models. *CoRR*, abs/2402.03300.

Tianhao Shen, Renren Jin, Yufei Huang, Chuang Liu, Weilong Dong, Zishan Guo, Xinwei Wu, Yan Liu, and Deyi Xiong. 2023. Large language model alignment: A survey. *Preprint*, arXiv:2309.15025.

Amogh Sheth, Andrew Lin, Yi Wen Huang, Biruk Assefa, and Yuhao Ge. 2026a. Pruning via causal attribution preserves reasoning in large language models. In *ICLR 2026 Workshop on Logical Reasoning of Large Language Models*.

Ivaxi Sheth, Sahar Abdelnabi, and Mario Fritz. 2024. Hypothesizing missing causal variables with LLMs. In *Causality and Large Models @NeurIPS 2024*.

Ivaxi Sheth, Bahare Fatemi, and Mario Fritz. 2025. CausalGraph2LLM: Evaluating LLMs for causal queries. In *Findings of the Association for Computational Linguistics: NAACL 2025*, pages 2076–2098, Albuquerque, New Mexico. Association for Computational Linguistics.

Ivaxi Sheth, Zhijing Jin, Bryan Wilder, Dominik Janzing, and Mario Fritz. 2026b. IV co-scientist: Multi-agent LLM framework for causal instrumental variable discovery. In *Fifth Conference on Causal Learning and Reasoning*.

Noah Shinn, Federico Cassano, Ashwin Gopinath, Karthik Narasimhan, and Shunyu Yao. 2023. Reflexion: language agents with verbal reinforcement learning. In *Advances in Neural Information Processing Systems*, volume 36, pages 8634–8652. Curran Associates, Inc.

Chathurangi Shyalika, Aryaman Sharma, Fadi El Kalach, Utkarshani Jaimini, Cory Henson, Ramy Harik, and Amit Sheth. 2026. Causaltrace: A neurosymbolic causal analysis agent for smart manufacturing. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 40, pages 40432–40439. AAAI.

Noah Siegel, Oana-Maria Camburu, Nicolas Heess, and Maria Perez-Ortiz. 2024. The probabilities also matter: A more faithful metric for faithfulness of free-text explanations in large language models. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 2: Short Papers)*, pages 530–546, Bangkok, Thailand. Association for Computational Linguistics.

Shashank Sonkar, Naiming Liu, MyCo Le, and Richard Baraniuk. 2024. MalAlgoQA: Pedagogical evaluation of counterfactual reasoning in large language models and implications for AI in education. In *Findings of the Association for Computational Linguistics: EMNLP 2024*, pages 15554–15567, Miami, Florida, USA. Association for Computational Linguistics.

Shovito Barua Soumma, Asiful Arefeen, Stephanie M. Carpenter, Melanie Hingle, and Hassan Ghasemzadeh. 2026. Counterfactual modeling with fine-tuned llms for health intervention design and sensor data augmentation. *IEEE Open Journal of Engineering in Medicine and Biology*, 7:232–240.

Saksham Sahai Srivastava. 2026. Causal intervention-based memory selection for long-horizon llm agents. *Preprint*, arXiv:2605.17641.

Alessandro Stolfo, Zhijing Jin, Kumar Shridhar, Bernhard Schölkopf, and Mrinmaya Sachan. 2023. A causal framework to quantify the robustness of mathematical reasoning with language models. In *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 545–561, Toronto, Canada. Association for Computational Linguistics.

Wentao Sun, João Paulo Nogueira, and Alonso Silva. 2025a. Structured thinking matters: Improving llms generalization in causal inference tasks. *Preprint*, arXiv:2505.18034.

Yijia Sun, Lyuzhou Chen, Xiangyu Wang, Lishan Yang, and Lihua Liu. 2025b. Llm-augmented causal discovery with order conflict resolution. In *2025 20th International Joint Symposium on Artificial Intelligence and Natural Language Processing (iSAI-NLP)*, pages 1–6.

Yuxi Sun, Aoqi Zuo, Haotian Xie, Wei Gao, Mingming Gong, and Jing Ma. 2026. FACT-E: Causality-inspired evaluation for trustworthy chain-of-thought reasoning. In *Findings of the Association for Computational Linguistics: ACL 2026*, pages 20275–20293, San Diego, California, United States. Association for Computational Linguistics.

Yuni Susanti and Michael Färber. 2025. Can LLMs leverage observational data? towards data-driven causal discovery with LLMs. *arXiv preprint arXiv:2504.10936*. Causal-NeSy Workshop at ESWC 2025.

Denis Sutter, Julian Minder, Thomas Hofmann, and Tiago Pimentel. 2025. The non-linear representation dilemma: Is causal abstraction enough for mechanistic interpretability? In *Advances in Neural Information Processing Systems*, volume 38, Main Conference, pages 149455–149503. Curran Associates, Inc.

Anand Swaroop, Akshat Nallani, Saksham Uboweja, Adiliia Uzdenova, Michael Nguyen, Kevin Zhu, Sunishchal Dev, Ashwinee Panda, Vasu Sharma, and Maheep Chaudhary. 2025. Frit: Using causal importance to improve chain-of-thought faithfulness. *Preprint*, arXiv:2509.13334.

Aaquib Syed, Can Rager, and Arthur Conmy. 2024. Attribution patching outperforms automated circuit discovery. In *Proceedings of the 7th BlackboxNLP Workshop: Analyzing and Interpreting Neural Networks for NLP*, pages 407–416, Miami, Florida, US. Association for Computational Linguistics.

MASAYUKI TAKAYAMA, Tadahisa OKUDA, Thong Pham, Tatsuyoshi Ikenoue, Shingo Fukuma, Shohei Shimizu, and Akiyoshi Sannai. 2025. Integrating large language models in causal discovery: A statistical causal approach. *Transactions on Machine Learning Research*.

Juanhe (TJ) Tan. 2023. Causal abstraction for chain-of-thought reasoning in arithmetic word problems. In *Proceedings of the 6th BlackboxNLP Workshop: Analyzing and Interpreting Neural Networks for NLP*, pages 155–168, Singapore. Association for Computational Linguistics.

Niket Tandon, Bhavana Dalvi Mishra, Keisuke Sakaguchi, Antoine Bosselut, and Peter Clark. 2019. WIQA: A dataset for "what if..." reasoning over procedural text. In *Proceedings of EMNLP-IJCNLP*.

Hao Tang, Darren Key, and Kevin Ellis. 2024. Worldcoder, a model-based llm agent: Building world models by writing code and interacting with the environment. In *Advances in Neural Information Processing Systems*, volume 37, pages 70148–70212. Curran Associates, Inc.

Ziyi Tang, Ruilin Wang, Weixing Chen, Yongsen Zheng, Zechuan Chen, Yang Liu, Keze Wang, Tianshui Chen, and Liang Lin. 2025. Towards causalgpt: A multi-agent approach for faithful knowledge reasoning via promoting causal consistency in llms. *Preprint*, arXiv:2308.11914.

Jin Tian and Judea Pearl. 2000. Probabilities of causation: Bounds and identification. *Annals of Mathematics and Artificial Intelligence*, 28:287–313.

Gilat Toker, Nitay Calderon, Ohad Amosy, and Roi Reichart. 2026. LIBERTy: A causal framework for benchmarking concept-based explanations of LLMs with structural counterfactuals. In *Findings of the Association for Computational Linguistics: ACL 2026*, pages 8149–8186, San Diego, California, United States. Association for Computational Linguistics.

Armin Toroghi, Faeze Moradi Kalarde, and Scott Sanner. 2026. Commonwhy: A dataset for evaluating entity-based causal commonsense reasoning in large language models. In *Proceedings of the 49th International ACM SIGIR Conference on Research and Development in Information Retrieval*, SIGIR '26, page 3418–3425, New York, NY, USA. Association for Computing Machinery.

Stelios Triantafyllou, Aleksa Sukovic, Yasaman Zolfimoselo, and Goran Radanovic. 2025. Counterfactual effect decomposition in multi-agent sequential decision making. In *Forty-second International Conference on Machine Learning*.

Ioannis Tsamardinos, Laura E. Brown, and Constantin F. Aliferis. 2006. The max-min hill-climbing bayesian network structure learning algorithm. *Machine Learning*, 65(1):31–78.

Polina Tsvilodub, Jan-Felix Klumpp, Amir Mohammadpour, Jennifer Hu, and Michael Franke. 2026. On emergent social world models — evidence for functional integration of theory of mind and pragmatic

reasoning in language models. In *Proceedings of the 64th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 37382–37420, San Diego, California, United States. Association for Computational Linguistics.

Miles Turpin, Julian Michael, Ethan Perez, and Samuel Bowman. 2023. Language models don't always say what they think: Unfaithful explanations in chain-of-thought prompting. In *Advances in Neural Information Processing Systems*, volume 36, pages 74952–74965. Curran Associates, Inc.

Jitesh Uikey. 2025. On the causal identifiability of chain-of-thought in language models. *TechRxiv*, 2025(1027).

Aniket Vashishtha, Qirun Dai, Hongyuan Mei, Amit Sharma, Chenhao Tan, and Hao Peng. 2026. Executable counterfactuals: Improving LLMs' causal reasoning through code. In *The Fourteenth International Conference on Learning Representations*.

Aniket Vashishtha, Abhinav Kumar, Atharva Pandey, Abbavaram Gowtham Reddy, Vineeth N. Balasubramanian, and Amit Sharma. 2024. Teaching transformers causal reasoning through axiomatic training. In *Causality and Large Models @NeurIPS 2024*.

Aniket Vashishtha, Abbavaram Gowtham Reddy, Abhinav Kumar, Saketh Bachu, Vineeth N. Balasubramanian, and Amit Sharma. 2025. Causal order: The key to leveraging imperfect experts in causal inference. In *The Thirteenth International Conference on Learning Representations*.

Marko Veljanovski and Zach Wood-Doughty. 2024. DoubleLingo: Causal estimation with large language models. In *Proceedings of the 2024 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies (Volume 2: Short Papers)*, pages 799–807, Mexico City, Mexico. Association for Computational Linguistics.

Vishal Verma, Sawal Acharya, Devansh Bhardwaj, Samuel Simko, Yongjin Yang, Anahita Haghighat, Dominik Janzing, Mrinmaya Sachan, Bernhard Schölkopf, and Zhijing Jin. 2025. Causal AI scientist: Facilitating causal data science with large language models. In *NeurIPS 2025 Workshop on CauScien: Uncovering Causality in Science*.

Sandra Wachter, Brent Mittelstadt, and Chris Russell. 2017. Counterfactual explanations without opening the black box: Automated decisions and the GDPR. *Harvard Journal of Law & Technology*, 31(2).

Guangya Wan, Yunsheng Lu, Yuqi Wu, Mengxuan Hu, and Sheng Li. 2025. Large language models for causal discovery: Current landscape and future directions. In *Proceedings of the Thirty-Fourth International Joint Conference on Artificial Intelligence, IJCAI-25*, pages 10687–10695. International Joint Conferences on Artificial Intelligence Organization. Survey Track.

Boxin Wang, Weixin Chen, Hengzhi Pei, Chulin Xie, Mintong Kang, Chenhui Zhang, Chejian Xu, Zidi Xiong, Ritik Dutta, Rylan Schaeffer, Sang T. Truong, Simran Arora, Manias Mazeika, Dan Hendrycks, Zinan Lin, Yu Cheng, Sanmi Koyejo, Dawn Song, and Bo Li. 2023. Decodingtrust: a comprehensive assessment of trustworthiness in gpt models. In *Proceedings of the 37th International Conference on Neural Information Processing Systems*, NIPS '23, Red Hook, NY, USA. Curran Associates Inc.

Guanzhi Wang, Yuqi Xie, Yunfan Jiang, Ajay Mandlekar, Chaowei Xiao, Yuke Zhu, Linxi Fan, and Anima Anandkumar. 2024. Voyager: An open-ended embodied agent with large language models. *Transactions on Machine Learning Research*.

Jingyao Wang, Peizheng Guo, Wenwen Qiang, Jiahuan Zhou, Huijie Guo, Changwen Zheng, and Hui Xiong. 2026a. Towards generalizable reasoning: Group causal counterfactual policy optimization for llm reasoning. *CoRR*, abs/2602.06475.

Nengbo Wang, Xiaotian Han, Jagdip Singh, Jing Ma, and Vipin Chaudhary. 2025a. CausalRAG: Integrating causal graphs into retrieval-augmented generation. In *Findings of the Association for Computational Linguistics: ACL 2025*, pages 22680–22693, Vienna, Austria. Association for Computational Linguistics.

Nengbo Wang, Tuo Liang, Vikash Singh, Chaoda Song, Van Yang, Yu Yin, Jing Ma, JAGDIP SINGH, and Vipin Chaudhary. 2026b. CausalRAG2: Hierarchical causal knowledge graph design for RAG. In *Forty-third International Conference on Machine Learning*.

Ruiqi Wang, Ruikang Liu, Runyu Chen, Haoxiang Suo, Zhiyi Peng, Zhuo Tang, and Changjian Chen. 2026c. Causaltad: Injecting causal knowledge into large language models for tabular anomaly detection. *Preprint*, arXiv:2602.07798.

Wenhui Wang, Furu Wei, Li Dong, Hangbo Bao, Nan Yang, and Ming Zhou. 2020. Minilm: Deep self-attention distillation for task-agnostic compression of pre-trained transformers. In *Advances in Neural Information Processing Systems*, volume 33, pages 5776–5788. Curran Associates, Inc.

Xinyue Wang, Kun Zhou, Wenyi Wu, Har Simrat Singh, Fang Nan, Songyao Jin, Aryan Philip, Saloni Patnaik, Hou Zhu, Shivam Singh, Parjanya Prashant, Qian Shen, and Biwei Huang. 2025b. Causal-copilot: An autonomous causal analysis agent. *Preprint*, arXiv:2504.13263.

Xinzhi Wang and Chunwei Liu. 2026. The table says otherwise: Testing LLMs with counterfactual relational data. In *VLDB 2026 Workshop: NOVAS*.

Yian Wang, Yuen Chen, Agam Goyal, and Hari Sundaram. 2026d. CausalDetox: Causal head selection and intervention for language model detoxification. In *Findings of the Association for Computational Linguistics: ACL 2026*, pages 11893–11914, San Diego, California, United States. Association for Computational Linguistics.

Yucheng Wang, Ziyang Chen, and Md Faisal Kabir. 2025c. Explaining fine-tuned LLMs via counterfactuals: A knowledge graph–driven framework. In *The First Structured Knowledge for Large Language Models Workshop*.

Yuzhe Wang, Yaochen Zhu, and Jundong Li. 2026e. Causalflip: A benchmark for llm causal judgment beyond semantic matching. In *Proceedings of the 32nd ACM SIGKDD Conference on Knowledge Discovery and Data Mining V.2*, KDD '26, page 9963–9972, New York, NY, USA. Association for Computing Machinery.

Zeyu Wang. 2024a. CausalBench: A comprehensive benchmark for evaluating causal reasoning capabilities of large language models. In *Proceedings of the 10th SIGHAN Workshop on Chinese Language Processing (SIGHAN-10)*, pages 143–151, Bangkok, Thailand. Association for Computational Linguistics.

Zeyu Wang. 2024b. CausalBench: A comprehensive benchmark for evaluating causal reasoning capabilities of large language models. In *Proceedings of the 10th SIGHAN Workshop on Chinese Language Processing*, pages 143–151. Association for Computational Linguistics.

Ziming Wang, Yinghua Yao, Changwu Huang, Ke Tang, and Xin Yao. 2026f. Case: Causal alignment and structural enforcement for improving chain-of-thought faithfulness. *Preprint*, arXiv:2607.18820.

Ziyu Wang, Elahe Khatibi, and Amir M. Rahmani. 2025d. Medcot-rag: Causal chain-of-thought rag for medical question answering. *Preprint*, arXiv:2508.15849.

Jason Wei, Xuezhi Wang, Dale Schuurmans, Maarten Bosma, Brian Ichter, Fei Xia, Ed Chi, Quoc V. Le, and Denny Zhou. 2022. Chain-of-thought prompting elicits reasoning in large language models. In *Advances in Neural Information Processing Systems*, volume 35, pages 24824–24837. Curran Associates, Inc.

Oscar William Lithgow-Serrano, Vani Kanjirangat, and Alessandro Antonucci. 2025. Causal understanding by LLMs: The role of uncertainty. In *Proceedings of the 2nd Workshop on Uncertainty-Aware NLP (UncertaiNLP 2025)*, pages 208–228, Suzhou, China. Association for Computational Linguistics.

Anpeng Wu, Kun Kuang, Minqin Zhu, Yingrong Wang, Yujia Zheng, Kairong Han, Baohong Li, Guangyi Chen, Fei Wu, and Kun Zhang. 2024a. Causality for large language models. *Preprint*, arXiv:2410.15319.

Jian Wu, Linyi Yang, Zhen Wang, Manabu Okumura, and Yue Zhang. 2025. Cofca: A step-wise counterfactual multi-hop qa benchmark. In *International Conference on Learning Representations*, volume 2025, pages 14631–14649.

Junda Wu, Tong Yu, Xiang Chen, Haoliang Wang, Ryan Rossi, Sungchul Kim, Anup Rao, and Julian McAuley. 2024b. DeCoT: Debiasing chain-of-thought for knowledge-intensive tasks in large language models via causal intervention. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 14073–14087, Bangkok, Thailand. Association for Computational Linguistics.

Linhao Wu, Yizhou Chen, Zhen Yang, Pengyu Xue, and Dan Hao. 2026. Causalrepair: Bridging the causality gap in large language model-based automated program repair via dual-slicing. *Preprint*, arXiv:2608.10613.

Zhengxuan Wu, Atticus Geiger, Thomas Icard, Christopher Potts, and Noah Goodman. 2023. Interpretability at scale: Identifying causal mechanisms in alpaca. In *Advances in Neural Information Processing Systems*, volume 36, pages 78205–78226. Curran Associates, Inc.

Yu Xia, Tong Yu, Zhankui He, Handong Zhao, Julian McAuley, and Shuai Li. 2024. Aligning as debiasing: Causality-aware alignment via reinforcement learning with interventional feedback. In *Proceedings of the 2024 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies (Volume 1: Long Papers)*, pages 4684–4695, Mexico City, Mexico. Association for Computational Linguistics.

Tian Xie, Tongxin Yin, Vaishakh Keshava, Xueru Zhang, and Siddhartha Reddy Jonnalagadda. 2026. Biascause: Evaluate socially biased causal reasoning of large language models. *Preprint*, arXiv:2504.07997.

Zhi Xu and Yun Fu. 2026. NoisyCausal: A benchmark for evaluating causal reasoning under structured noise. In *Proceedings of the 64th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 39500–39513, San Diego, California, United States. Association for Computational Linguistics.

Khurram Yamin, Gaurav Rohit Ghosal, and Bryan Wilder. 2025a. LLMs struggle to perform counterfactual reasoning with parametric knowledge. In *ICML 2025 Workshop on Scaling Up Intervention Models*.

Khurram Yamin, Gaurav Rohit Ghosal, and Bryan Wilder. 2025b. LLMs struggle to perform counterfactual reasoning with parametric knowledge. In *ICML 2025 Workshop on Scaling Up Intervention Models*.

Khurram Yamin, Shantanu Gupta, Gaurav Rohit Ghosal, Zachary Chase Lipton, and Bryan Wilder. 2025c. Failure modes of LLMs for causal reasoning on narratives. In *ICML 2025 Workshop on Scaling Up Intervention Models*.

Zirui Yan, Dennis Wei, Dmitriy A Katz, Prasanna Sattigeri, and Ali Tajer. 2026. Multi-component causal tracing in large language models. In *Proceedings of the 64th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 3398–3418, San Diego, California, United States. Association for Computational Linguistics.

Bo Yang, Jiaxian Guo, Yusuke Iwasawa, and Yutaka Matsuo. 2025. Large language models as theory of mind aware generative agents with counterfactual reflection. *Preprint*, arXiv:2501.15355.

Shuai Yang, Qi Yang, Luoxi Tang, Yuqiao Meng, Nancy Guo, Jeremy Blackburn, and Zhaohan Xi. 2026. On the eligibility of LLMs for counterfactual reasoning: A decompositional study. In *The Fourteenth International Conference on Learning Representations*.

Jin Yao, Eli Chien, Minxin Du, Xinyao Niu, Tianhao Wang, Zezhou Cheng, and Xiang Yue. 2024. Machine unlearning of pre-trained large language models. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 8403–8419, Bangkok, Thailand. Association for Computational Linguistics.

Shunyu Yao, Dian Yu, Jeffrey Zhao, Izhak Shafran, Thomas L. Griffiths, Yuan Cao, and Karthik Narasimhan. 2023a. Tree of thoughts: deliberate problem solving with large language models. In *Proceedings of the 37th International Conference on Neural Information Processing Systems*, NIPS '23, Red Hook, NY, USA. Curran Associates Inc.

Shunyu Yao, Jeffrey Zhao, Dian Yu, Nan Du, Izhak Shafran, Karthik R Narasimhan, and Yuan Cao. 2023b. React: Synergizing reasoning and acting in language models. In *The Eleventh International Conference on Learning Representations*.

Xiaotian Ye, Xiaohan Wang, Mengqi Zhang, and Shu Wu. 2026. On the hidden costs of counterfactual knowledge training in llm unlearning. *Preprint*, arXiv:2605.27083.

Wei Jie Yeo, Ranjan Satapathy, and Erik Cambria. 2025. Towards faithful natural language explanations: A study using activation patching in large language models. In *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing*, pages 10425–10447, Suzhou, China. Association for Computational Linguistics.

Jialin Yu, Yuxiang Zhou, Haoxuan Li, Junchi Yu, Mengyue Yang, Yulan He, Nevin L. Zhang, Philip Torr, and Ricardo Silva. 2026a. Causal fine-tuning under latent confounded shift. In *Forty-third International Conference on Machine Learning*.

Longxuan Yu, Delin Chen, Siheng Xiong, Qingyang Wu, Dawei Li, Zhikai Chen, Xiaoze Liu, and Liangming Pan. 2025a. CausalEval: Towards better causal reasoning in language models. In *Proceedings of the 2025 Conference of the Nations of the Americas Chapter of the Association for Computational Linguistics: Human Language Technologies (Volume 1: Long Papers)*, pages 12512–12540, Albuquerque, New Mexico. Association for Computational Linguistics.

Xiangning Yu, Yuwei Guo, Yuqi Hou, Xiao Xue, and Qun Ma. 2026b. CAMO: An agentic framework for automated causal discovery from micro behaviors to macro emergence in LLM agent simulations. In *Findings of the Association for Computational Linguistics: ACL 2026*, pages 24447–24479, San Diego, California, United States. Association for Computational Linguistics.

Xiangning Yu, Zhuohan Wang, Linyi Yang, Haoxuan Li, Anjie Liu, Xiao Xue, Jun Wang, and Mengyue Yang. 2025b. Causal sufficiency and necessity improves chain-of-thought reasoning. In *Advances in Neural Information Processing Systems*, volume 38, Main Conference, pages 126109–126141. Curran Associates, Inc.

Kerem Zaman and Shashank Srivastava. 2025. A causal lens for evaluating faithfulness metrics. In *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing*, pages 29425–29449, Suzhou, China. Association for Computational Linguistics.

Lei Zan, Keli Zhang, Ruichu Cai, and Lujia Pan. 2026. Cama: Enhancing mathematical reasoning in large language models with causal knowledge. In *Proceedings of the AAAI Symposium Series, 40(41), 34566–34574*.

Khadija Zanna and Akane Sano. 2025. Fairness-driven LLM-based causal discovery with active learning and dynamic scoring. *arXiv preprint arXiv:2503.17569*.

Matej Zečević, Moritz Willig, Devendra Singh Dhami, and Kristian Kersting. 2023. Causal parrots: Large language models may talk causality but are not causal. *Transactions on Machine Learning Research*.

Cheng Zhang, Stefan Bauer, Paul Bennett, Jiangfeng Gao, Wenbo Gong, Agrin Hilmkil, Joel Jennings, Chao Ma, Tom Minka, Nick Pawlowski, and James Vaughan. 2023. Understanding causality with large language models: Feasibility and opportunities. *Preprint*, arXiv:2304.05524.

Congzhi Zhang, Linhai Zhang, Jialong Wu, Yulan He, and Deyu Zhou. 2025a. Causal prompting: debiasing large language model prompting based on front-door adjustment. In *Proceedings of the Thirty-Ninth AAAI Conference on Artificial Intelligence and Thirty-Seventh Conference on Innovative Applications of Artificial Intelligence and Fifteenth Symposium on Educational Advances in Artificial Intelligence*, AAAI'25/IAAI'25/EAAI'25. AAAI Press.

Fred Zhang and Neel Nanda. 2024. Towards best practices of activation patching in language models: Metrics and methods. In *International Conference on Learning Representations*, volume 2024, pages 1651–1678.

Luhao Zhang, Xinyu Zhang, Linmei Hu, Dandan Song, and Liqiang Nie. 2025b. Dually self-improved counterfactual data augmentation using large language model. In *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 5216–5227, Vienna, Austria. Association for Computational Linguistics.

Mengdi Zhang, Goh Kai Kiat, Peixin Zhang, Jun Sun, Lin Xin Rose, and Hongyu Zhang. 2025c. LLMScan: Causal scan for LLM misbehavior detection. In *Proceedings of the 42nd International Conference on Machine Learning*, volume 267 of *Proceedings of Machine Learning Research*, pages 75253–75282. PMLR.

Tian Zhang, Yiwei Xu, Juan Wang, Keyan Guo, Xiaoyang Xu, Bowen Xiao, Quanlong Guan, Jinlin Fan, Jiawei Liu, Zhiquan Liu, and Hongxin Hu. 2026a. Agentsentry: Mitigating indirect prompt injection in llm agents via temporal causal diagnostics and context purification. *Preprint*, arXiv:2602.22724.

Xiaohui Zhang, Zequn Sun, Chengyuan Yang, Yaqin Jin, Yazhong Zhang, and Wei Hu. 2026b. Actmem: Bridging the gap between memory retrieval and reasoning in llm agents. *Preprint*, arXiv:2603.00026.

Yanxi Zhang, Xin Cong, Zhong Zhang, Xiao Liu, Dongyan Zhao, and Yesai Wu. 2025d. Synergizing large language models and theory for human-like causal reasoning. In *NeurIPS 2025 First Workshop on LLM Persona Modeling*.

Yuanjun Zhang and Mourad Oussalah. 2026. Causal evidence extraction and triangulation in crisis reports using large language models: A reliefweb-based study. In *Findings of the Association for Computational Linguistics: ACL 2026*, page 32478–32491. Association for Computational Linguistics.

Zhenliang Zhang, Junzhe Zhang, Xinyu Hu, Huixuan Zhang, and Xiaojun Wan. 2025e. Exploring causal effect of social bias on faithfulness hallucinations in large language models. In *Proceedings of the 34th ACM International Conference on Information and Knowledge Management*, CIKM '25, page 4293–4303, New York, NY, USA. Association for Computing Machinery.

Bo Zhao, Yinghao Zhang, Ziqi Xu, Yongli Ren, Xiuzhen Zhang, Renqiang Luo, Zaiwen Feng, and Feng Xia. 2025. Unbiased reasoning for knowledge-intensive tasks in large language models via conditional front-door adjustment. *Preprint*, arXiv:2508.16910.

Xiaoling Zhou, Mingjie Zhang, Zhemg Lee, YUNCHENG HUA, chengli xing, Wei Ye, Flora Salim, and Shikun Zhang. 2025. Boosting resilience of large language models through causality-driven robust optimization. In *Advances in Neural Information Processing Systems*, volume 38, Main Conference, pages 55821–55845. Curran Associates, Inc.

Yao Zhou, Zeen Song, Wenwen Qiang, Fengge Wu, Shuyi Zhou, Changwen Zheng, and Hui Xiong. 2026. Causal front-door adjustment for robust jailbreak attacks on llms. *Preprint*, arXiv:2602.05444.

Haohao Zhu, Tingting Li, Peng He, and Jiayu Zhou. 2025. Enhancing automated grading in science education through llm-driven causal reasoning and multimodal analysis. In *Proceedings of the Thirty-Fourth International Joint Conference on Artificial Intelligence, IJCAI-25*, pages 10352–10360. International Joint Conferences on Artificial Intelligence Organization. Human-Centred AI.

Kaijie Zhu, Jindong Wang, Jiaheng Zhou, Zichen Wang, Hao Chen, Yidong Wang, Linyi Yang, Wei Ye, Yue Zhang, Neil Zhenqiang Gong, and Xing Xie. 2024. Promptrobust: Towards evaluating the robustness of large language models on adversarial prompts. *Preprint*, arXiv:2306.04528.

A Causality Background

In this section, we give an overview of the background in causal inference. We introduce the key concepts and notations used throughout this survey. First, we define causal graphs and structural causal models, following the framework of Pearl (2009), then we move to define individual, conditional and average treatment effects using the potential outcomes framework of Rubin (1974). Finally, we define counterfactual inference, which concerns the question of what would have happened under a different intervention given what was actually observed.

Definition 1 (Causal Graph). A causal graph is a directed acyclic graph (DAG) $G = (V, E)$, where $V = \{V_1, \ldots, V_d\}$ is a set of random variables and $E \subseteq V \times V$ is a set of directed edges representing direct causal relationships. An edge $V_i \to V_j$ indicates that V_i is a direct cause of V_j relative to the variables represented in G. The parents of V_i, denoted by $\mathrm{Pa}_G(V_i)$, are the variables that directly influence V_i.

Definition 2 (Structural Causal Model). A structural causal model (SCM) is a tuple $\mathcal{M} = (\mathbf{U}, \mathbf{V}, \mathcal{F}, P_{\mathbf{U}})$, where $\mathbf{U}$ denotes exogenous variables, $\mathbf{V} = \{V_1, \ldots, V_d\}$ denotes endogenous variables, $\mathcal{F} = \{f_1, \ldots, f_d\}$ is a collection of structural functions, and $P_{\mathbf{U}}$ is a joint probability distribution over the exogenous variables. Each endogenous variable V_i is determined by a structural equation

$$V_i = f_i\big(\mathrm{Pa}_G(V_i), U_i\big),$$

where U_i denotes the exogenous inputs to f_i. The structural equations specify the causal mechanisms generating the endogenous variables. An intervention $\mathrm{do}(T = t)$ replaces the structural equation for T by $T = t$, leaving the remaining structural equations unchanged.

Definition 3 (Individual Treatment Effect). Let $T \in \{0, 1\}$ denote a binary treatment and let $Y_i(t)$ denote the potential outcome of individual i under intervention $T = t$. The individual treatment effect (ITE) is defined as

$$\tau_i = Y_i(1) - Y_i(0).$$

The ITE quantifies the causal effect of changing the treatment from 0 to 1 for individual i. Since only one potential outcome is observed for each individual, the ITE is generally not directly identifiable from observed data.

Definition 4 (Conditional Average Treatment Effect). Let X denote a vector of pretreatment covariates. The conditional average treatment effect (CATE) at $X = x$ is defined as

$$\tau(x) = \mathbb{E}\big[Y(1) - Y(0) \mid X = x\big].$$

Equivalently,

$$\tau(x) = \mathbb{E}[Y(1) \mid X = x] - \mathbb{E}[Y(0) \mid X = x].$$

The CATE quantifies the average causal effect among individuals whose covariates satisfy $X = x$, allowing treatment effects to vary across covariate-defined subpopulations.

Definition 5 (Average Treatment Effect). The average treatment effect (ATE) is the expected difference between the potential outcomes under treatment and control:

$$\mathrm{ATE} = \mathbb{E}[Y(1) - Y(0)].$$

Equivalently,

$$\mathrm{ATE} = \mathbb{E}[Y(1)] - \mathbb{E}[Y(0)] = \mathbb{E}_X[\tau(X)].$$

The ATE quantifies the average causal effect of the treatment across the target population.

Definition 6 (Counterfactual Inference). Let $\mathcal{M}$ be a structural causal model, let $E = e$ denote observed evidence, and let Y_t denote the outcome that would result from the hypothetical intervention $\mathrm{do}(T = t)$. Counterfactual inference, in Pearl's structural causal framework, consists of evaluating

$$P_{\mathcal{M}}(Y_t = y \mid E = e).$$

This quantity represents the probability that the outcome would equal y under the hypothetical intervention $T = t$, conditional on the observed evidence $E = e$.

Counterfactual inference follows three steps:

1. **Abduction:** Update the distribution of exogenous variables using the evidence,

$$P_{\mathbf{U}}(\mathbf{u} \mid E = e).$$

2. **Action:** Construct the intervened model $\mathcal{M}_t$ by replacing the structural equation for T by $T = t$.

3. **Prediction:** Compute the distribution of Y in $\mathcal{M}_t$ under the updated distribution of the exogenous variables:

$$\begin{aligned} &P_{\mathcal{M}}(Y_t = y \mid E = e) \\ &= \int \mathbb{I}\{Y_{\mathcal{M}_t}(\mathbf{u}) = y\}\, dP_{\mathbf{U}}(\mathbf{u} \mid E = e). \end{aligned}$$

   assuming the structural equations admit a unique solution.

Unlike interventional inference, counterfactual inference conditions on factual evidence before evaluating a hypothetical intervention.

B Detailed Taxonomy

The main paper focuses on the individual topics of the relationship between causality and LLMs. In Figure 3 of this appendix shows the complete taxonomy. The taxonomy is organized into five main areas: causality and system design, causality and capability, causality and trustworthiness, applications, and evaluation. Each area is then divided into the topics covered in the main paper, and each topic is shown with the papers and sections where they are discussed.

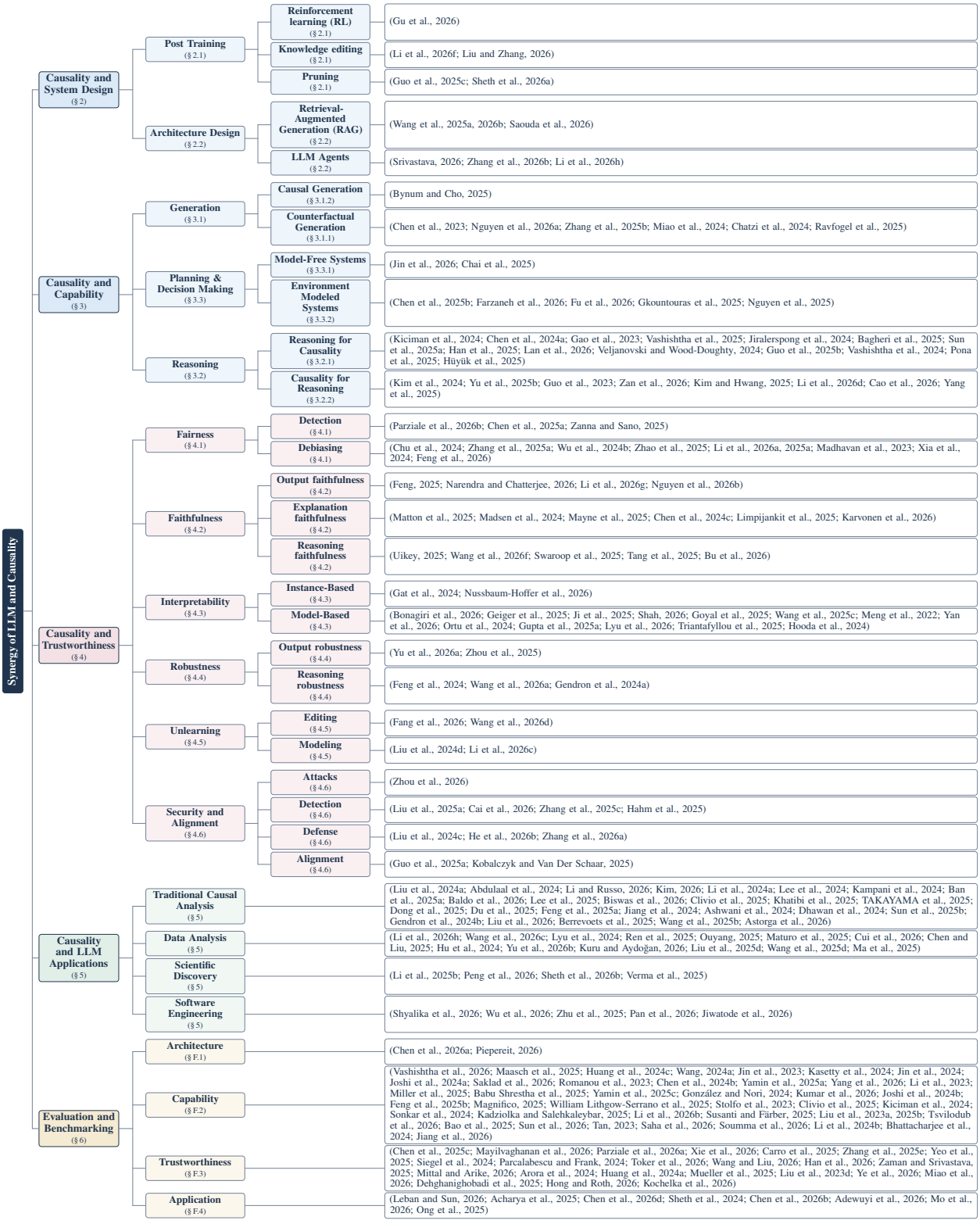


Figure 3: Taxonomy for synergy between LLM and causality.

C Detailed Comparison with Existing Surveys

Prior surveys on causality and LLMs typically address only one direction of the relationship: either how LLMs perform causal tasks, or how causal methods improve LLMs. Table 2 evaluates each survey against both directions and against the five areas of our taxonomy. We consider an area covered only when a survey discusses multiple works on that area in depth, rather than citing them in passing and under this criterion, no prior survey covers all five areas, which is the gap our survey addresses.

Study	Focus	LLM for Causality	Causality for LLMs	System Design	Capability	Trustworthiness	Apps.	Eval.
Zhang et al. (2023)	Causal feasibility and opportunities	⚠	✗	✗	✗	✗	✗	✗
Wu et al. (2024a)	Causality across the LLM lifecycle	⚠	✓	⚠	✓	✓	⚠	⚠
Cui et al. (2024)	Commonsense causality and reasoning	✗	✗	✗	✗	✗	✗	⚠
Ma (2025)	LLMs for causal inference	✓	✗	✗	⚠	✗	⚠	⚠
Liu et al. (2025c)	Causal–LLM collaboration, reasoning, trustworthiness, and multimodality	✓	✓	✗	✓	✓	⚠	⚠
Yu et al. (2025a)	Causal reasoning review and evaluation	✓	✗	✗	⚠	✗	⚠	⚠
Cui et al. (2025)	Causal uncertainty	⚠	✗	✗	⚠	✗	✗	✗
Bazgir et al. (2025)	Causal multi-agent LLM systems	✓	✗	✗	⚠	✗	⚠	⚠
Wan et al. (2025)	LLMs for causal discovery	✓	✗	✗	⚠	✗	⚠	✗
Frauen et al. (2026)	Causal LLM development and evaluation	✗	✓	⚠	✗	✗	✗	⚠
Li et al. (2025c)	Causal reasoning enhancement	✓	✗	✗	⚠	✗	⚠	⚠
Ours	**Causal LLM methods and evaluation**	✓	✓	✓	✓	✓	✓	✓

Table 2: Coverage of related surveys and studies, assessed against the taxonomy. In the five coverage areas, ✓: developed coverage of most branches of the area; ⚠: developed coverage limited to one or two branches; ✗: scattered citations or no developed coverage. **System Design**: causal methods for post-training and model design; **Capability**: generation, reasoning, and planning; **Trustworthiness**: fairness and bias, faithfulness, interpretability, robustness, unlearning, and security/alignment; **Apps.**: causal analysis, data analysis, scientific discovery, and software engineering; **Eval.**: Evaluation covers system design, capability, trustworthiness, and applications.

D Benchmarks, Datasets, and Metrics

Evaluation resources for causal reasoning in LLMs have been scattered across multiple independent works, making it hard to compare and evaluate them. Hence, we compile these resources into two tables, Table 3 and Table 1, to help readers compare them. The Table 3 lists benchmarks, datasets, and metrics by causal tasks. Readers can use this table to determine which resource is relevant to the causal task of interest, and what metrics are used to evaluate it. Table 1 regroups benchmarks and datasets by the LLM aspect they evaluate: system design, capability, trustworthiness, and applications. Each item in the tables cites the paper that introduced it.

E Application with More Details

In this section, we discuss in more detail the applications of LLM and causality. This is an expansion of our overview in Section 5. We cover the applications of data analysis, scientific discovery and software engineering.

Data Analysis Causal LLMs are used for various data analysis tasks. For anomaly detection they use the causal knowledge for tabular anomaly detection (Wang et al., 2026c). Causal LLMs are also used for analysis tasks involving sentiment analysis (Lyu et al., 2024; Ren et al., 2025; Ouyang, 2025), bankruptcy analysis (Maturo et al., 2025), time-series analysis (Cui et al., 2026; Biswas et al., 2026), sensory data analysis (Chen and Liu, 2025; Hu et al., 2024), and social emergence analysis (Yu et al., 2026b). Apart from analysis they are used in information retrieval (Jeong et al., 2026; Zhang and Oussalah, 2026; Naduvilakandy et al., 2025), recommendation systems (Kuru and Aydoğan, 2026; Liu et al., 2025d), question-answering (Wang et al., 2025d) and source fact-checking (Ma et al., 2025).

Scientific Discovery AI scientists in which AI systems, including LLMs, perform scientific discovery tasks have recently emerged. Recent studies have explored the causal capabilities of LLMs for experimental design (Li et al., 2025b), hypothesis generation (Peng et al., 2026), and instrumental variable discovery (Sheth et al., 2026b).

Software Engineering Causal LLMs are utilized to assist with software engineering tasks such as system analysis that uses causality identification methods to find key features in a programs (Shyalika et al., 2026; Wu et al., 2026), using causal reasoning for evaluation and optimization (Zhu et al., 2025; Pan et al., 2026), or using causal induction to reverse engineering video game description language (VGDL) rules from gameplay traces (Jiwatode et al., 2026).

F Evaluation with More Details

In this section, we discuss in more detail the evaluation of causal methods for LLMs. This is an expansion of our overview in Section 6. We cover the evaluation of system design, capability, trustworthiness, and applications as per our taxonomy. For each of them, we discuss the benchmarks and datasets that have been proposed or adapted, and how studies use interventions and counterfactuals to evaluate model behavior. The resources discussed in this section can be found in Table 3 and Table 1.

F.1 Architecture

To evaluate the impact of post-training methods on LLM's causal inference abilities CAUSALGYM (Chen et al., 2026a) evaluates the extent to which post-training methods enhance LLMs' capacity for causal inference. CausalGym comprises of seven core causal tasks for training and five diverse test sets. Using this dataset, they evaluate five post-training approaches: SFT, DPO, KTO, PPO, and GRPO.

For cross-model activation transfer, existing methods evaluate the representational similarity of the transferred activations and do not establish the causal influence (Piepereit, 2026). To bridge the gap, the authors intervene on the target model's hidden states injecting the source model's activations and a negative injection unrelated to the source model. They compare both the outputs to evaluate the causal effect of source model's activation transfer on the target model.

F.2 Capability

In this section, we highlight the different evaluation methods for evaluating the different capabilities of LLMs that are reasoning, generation, and planning.

To evaluate causal reasoning capabilities of LLMs some studies have relied on benchmarks and datasets that were originally designed for traditional causal tasks such as BNLearn (Scutari, 2010b) and Tübingen (Mooij et al., 2016). In some cases authors had to adapt these datasets to suit LLMs or their tasks (Kasetty et al., 2024; Clivio et al., 2025; Kiciman et al., 2024). However, various contributions have been made for datasets and benchmarks for causality in LLMs such as Cladder (Jin et al., 2023), Corr2Cause Jin et al. (2024),

COLD (Joshi et al., 2024a), CLEAR (Chen et al., 2024b), CRAB (Romanou et al., 2023), CausalBench (Wang, 2024b), RECITE (Saklad et al., 2026) that enable the evaluation of LLMs on different causal reasoning aspects such as formal, real-world, long-horizon, graph-based or certain novel tasks like MalAlgoQA (Sonkar et al., 2024) and CLOMO (Huang et al., 2024d). A comprehensive list of benchmarks can be found in Table 3.

A common focus while evaluating the LLMs for causal reasoning is to evaluate on the trustworthiness of their performed reasoning (Feng et al., 2025b; Kadziolka and Salehkaleybar, 2025; Kumar et al., 2026; William Lithgow-Serrano et al., 2025; Yang et al., 2026; Yamin et al., 2025c; Li et al., 2026b) and the extent of their reasoning capability (Vashishtha et al., 2026; Joshi et al., 2024b; Magnifico, 2025; Susanti and Färber, 2025; Miller et al., 2025; Li et al., 2023; Yamin et al., 2025b; Liu et al., 2023a; González and Nori, 2024; Liu et al., 2025b; Babu Shrestha et al., 2025).

Some evaluations for other reasoning capabilities have used causality as a medium for evaluation. For example, to evaluate the robustness of LLM in mathematical reasoning, a study develops a causal model of the the mathematical factors such as the surface form of the question, the operands, the math operators and the solution they study the behavior of LLMs to direct interventions in the input space (Stolfo et al., 2023). Similarly, to study how causal quantities compose across a graph or how the behavior changes when the composition or decomposition of causal measure change with graph complexity a study formalizes compositional causal reasoning CCR as the ability to infer the composition of local reasoning on global level and the decomposition of global level on local reasoning (Maasch et al., 2025). They define a framework which generates ground-truth for local components and its global composition and use that to evaluate the consistency of reasoning against ATE and PNS for composition and decomposition tasks. Alternatively, a study uses causal-mechanistic experiments to understand LLM's ToM abilities comprehensively (Tsvilodub et al., 2026).

Similarly, many studies have evaluated CoT reasoning with help of causality to understand its functioning and reliability (Bao et al., 2025; Sun et al., 2026; Tan, 2023; Saha et al., 2026).

Evaluations for generation capability of LLMs evaluate the extent of LLM's counterfactual generation on NLU tasks (Soumma et al., 2026; Li et al., 2024b), or data generation (Jiang et al., 2026).

A study investigates the effect of reasoning and semantic retrieval in CoT-reasoning (Kim et al., 2024). The authors compare the performance of LLM for reasoning questions based on RAG for retrieval-based setting, graph-inference for reasoning-based setting and proposed KG-based random-walk prompting framework for combination-setting. Their results reveal that incorporating causal relationships and structured reasoning can improve reasoning in LLMs.

F.3 Trustworthiness

A research highlights the lack of evaluating the spurious correlations in LLM semantics, existing studies detect them with co-occurrence statistics from training which are unavailable for LLMs. To assess the gap, LCD (Miao et al., 2026) identifies spurious correlations within the help of statistics encoded in the LLM parameters leveraging causal discovery. Based on these, they further introduce Spurious Correlation Evaluator (SCE) to assess the robustness using noisy data containing the identified spurious correlations.

It is important to evaluate the effectiveness of different methods to improve safety in LLMs for which researchers have evaluated different AI safety elements with a causal perspective.

Evaluation strategies in fairness focuses on counterfactual fairness (Parziale et al., 2026a; Mayilvaghanan et al., 2026), cognitive bias (Carro et al., 2025), social bias (Xie et al., 2026; Zhang et al., 2025e), and gender bias (Chen et al., 2025c). Faithfulness evaluations try to establish the trustworthiness of LLM reasoning based on concepts (Toker et al., 2026), SCE (Dehghanighobadi et al., 2025), CfSim (Hong and Roth, 2026), tabular reasoning (Wang and Liu, 2026), CoT reasoning (Parcalabescu and Frank, 2024) and large reasoning models (LRM) (Han et al., 2026). Similarly, interpretability evaluation focuses on understanding the effect of new methods causally on models architectures (Mueller et al., 2025; Arora et al., 2024; Huang et al., 2024a). Robustness and security rely on counterfactual generation or counterfactual analysis to establish the performance of LLMs in novel scenarios (Kochelka et al., 2026; Ye et al., 2026; Liu et al., 2023d).

F.4 Application

To understand the behavior of LLMs researchers have used causality as a medium for their evaluations. With the rising adoption of LLMs as AI scientists, researchers have proposed benchmarks to evaluate LLMs for their causal capabilities as Causal AI Scientists. Some studies focus on the causal reasoning capability of AI Scientists (Leban and Sun, 2026; Acharya et al., 2025; Chen et al., 2026d), while others propose novel tasks (Sheth et al., 2024). To ensure safe LLM reasoning for medical recommendations, studies have leveraged counterfactuals to generate benchmarks and datasets focused on robustness (Chen et al., 2026b), fairness (Adewuyi et al., 2026), and faithfulness (Mo et al., 2026).

Alternatively, a study investigated the use of forward counterfactual generation as a means to benchmark LLMs to anticipate future market developments (Ong et al., 2025).

Task	Benchmarks	Datasets	Metrics
Causal Reasoning	CLadder (Jin et al., 2023); CLEAR (Chen et al., 2024b); CausalGraph2LLM (Sheth et al., 2025); CausalEval (Yu et al., 2025a); METER (Li et al., 2026e); CausalPitfalls (Du et al., 2026); Causal Chains, Natural Word Chain, and Intuitive Physics tasks (Zečević et al., 2023); CausalDS (Leban and Sun, 2026); CauSciBench (Acharya et al., 2025); CauGym (Chen et al., 2026a); CaLM (Chen et al., 2024a); NoisyCausal (Xu and Fu, 2026); CausalProbe-2024 (Chi et al., 2024); AC-Bench (Zhang et al., 2025d); CRAB (Romanou et al., 2023); CausalBench (Wang, 2024a); CausalFlip (Wang et al., 2026e); QRData (Liu et al., 2024b)	ExpliCa (Miliani et al., 2025); Corr2Cause (Jin et al., 2024); CauSim (Astorga et al., 2026); QRText (Liu et al., 2024b); CausalityCheck-CP (Li et al., 2026b); COPA (Roemmele et al., 2011); e-CARE (Du et al., 2022); WIQA (Tandon et al., 2019); CommonWhy (Toroghi et al., 2026)	DoVerifier (He et al., 2026a)
Causal Inference	CausalReasoning Benchmark (Sawarni et al., 2026); Missing-causal-variable benchmark (Sheth et al., 2024)	CandyParty (González and Nori, 2024); Null-contingency medical scenarios (Carro et al., 2025)	ATE error (Rubin, 1974); CATE / ITE error (Shalit et al., 2017); PEHE (Hill, 2011) / standardized PEHE (Bynum and Cho, 2025); Probability of Necessity and Sufficiency (Tian and Pearl, 2000); Probability of Necessity (Tian and Pearl, 2000); Probability of Sufficiency (Tian and Pearl, 2000); Compositional ATE / PNS error (Maasch et al., 2025); Causal identification accuracy (Sawarni et al., 2026); Causal estimation error (Sawarni et al., 2026); Mediation Influence Score (Sheth et al., 2024)
Causal Discovery	CausalGraphBench (Babakov et al., 2025); Latent-Confounder benchmark (Ding et al., 2026)	bnlearn Bayesian-network repository (Scutari, 2010a); Sachs (Sachs et al., 2005); Tübingen Cause–Effect Pairs (Mooij et al., 2016); ER and scale-free synthetic DAGs (Erdős and Rényi, 1959; Barabási and Albert, 1999)	Structural Hamming Distance (Tsamardinos et al., 2006); Normalized Hamming Distance (Kumar et al., 2026); Structural Intervention Distance (Peters and Bühlmann, 2015); Adjacency precision / recall / F1; Edge FPR / FDR / TPR; Edge AUROC / AUPRC; Causal-edge calibration (Kumar et al., 2026); Causal DAG construction score (Li et al., 2026g)

Continued on the next page

Table 3 continued from the previous page

Task	Benchmarks	Datasets	Metrics
Counterfactual Reasoning	CounterBench (Chen et al., 2026c); CofCA (Wu et al., 2025); CounterLogic (Joishy et al., 2026); MalAlgoQA (Sonkar et al., 2024); RECALL (Liu et al., 2023c); ContraTable (Wang and Liu, 2026); RFEval (Han et al., 2026); MamaBench (Adewuyi et al., 2026); MedEinst (Chen et al., 2026b); MEDCOUNTERFACT (Mo et al., 2026)	CRASS (Frohberg and Binder, 2022); Executable-Code Counterfactuals (Vashishtha et al., 2026); Counterfactual-simulatability sets (Hong and Roth, 2026); DailyDilemmas (Chiu et al., 2025)	Counterfactual-query accuracy; Factual / counterfactual inconsistency (González and Nori, 2024); Algorithm / malgorithm identification (Sonkar et al., 2024); Counterfactual performance gap; Bias Trap Rate (Chen et al., 2026b); Counterfactual evidence adherence (Mo et al., 2026)
Counterfactual Generation	CEval (Nguyen et al., 2024); FIN-FORCE (Ong et al., 2025); CHIVE (Karvonen et al., 2026)	CLOMO (Huang et al., 2024c)	Counterfactual validity / target success (Wachter et al., 2017); Proximity / minimality (Wachter et al., 2017); Plausibility / naturalness; Diversity (Mothilal et al., 2020); Self-Evaluation Score (Huang et al., 2024c); Excess Gower distance (Mayne et al., 2025)
Causal Generation	SD-SCM causal-effect benchmark (Bynum and Cho, 2025)		
Planning & Decision Making	CausalGame (Chen et al., 2026d)	SymFail (Luan et al., 2026); CAR planted-SCM cases (Shah, 2026)	Failure reproduction rate (Luan et al., 2026); Conditional causal-repair rate (Luan et al., 2026); Single-step contrastive causal effect (Shah, 2026); Shapley causal-attribution recovery (Shah, 2026)

Table 3: Benchmarks, datasets, and metrics for evaluating the causal abilities of large language models, grouped by causal task. A benchmark is a task suite with a fixed protocol, a dataset is a collection of examples for training or testing, and a metric is a score for model outputs. Each entry cites the paper that introduced it.